DEI
DEPARTAMENTO
DE ENGENHARIA INFORMÁTICA
TÉCNICO LISBOA

Sistemas de Informação



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Agenda

- Enquadramento de **Sistemas de Informação** no currículo de Eng. Informática no IST
- Quem Somos
- Diagrama das Subáreas de **Sistemas de Informação**
- Saídas Profissionais em **Sistemas de Informação**

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Enquadramento de Sistemas de Informação no currículo de Eng. Informática no IST

A Área de Sistemas de Informação do DEI tem como objetivos científicos e pedagógicos principais:

(i) Ensinar e investigar os fundamentos para lidar com as empresas enquanto organizações, vistas como sistemas complexos dos quais os sistemas de informação fazem parte; e

(ii) Ensinar e investigar os fundamentos para conceber e operar sistemas informáticos onde a complexidade da informação e processos é elevada, quer em função da quantidade de dados a processar e interpretar, quer da dimensão das organizações e comunidades que os utilizam.

Para os atingir, é requerida a prática e conhecimentos especializados para o desenvolvimento de atividades de investigação e inovação em engenharia de sistemas de informação.

Sistemas de Informação

Conformidade com o *Body of Knowledge* do CS2013

Information Management (IM) is primarily concerned with the capture, digitization, representation, organization, transformation, and presentation of information; algorithms for efficient and effective access and updating of stored information; data modeling and abstraction; and physical file storage techniques. The student needs to be able to develop conceptual and physical data models, determine which IM methods and techniques are appropriate for a given problem, and be able to select and implement an appropriate IM solution that addresses relevant design concerns including scalability, accessibility and usability

Sistemas de Informação

Conformidade com o *Body of Knowledge* do CS2013

IM. Information Management (1 Core-Tier1 hour; 9 Core-Tier2 hours)

	Core-Tier1 hours	Core-Tier2 hours	Includes Electives
IM/Information Management Concepts	1	2	N
IM/Database Systems		3	Y
IM/Data Modeling		4	N
IM/Indexing			Y
IM/Relational Databases			Y
IM/Query Languages			Y
IM/Transaction Processing			Y
IM/Distributed Databases			Y
IM/Physical Database Design			Y
IM/Data Mining			Y
IM/Information Storage And Retrieval			Y
IM/MultiMedia Systems			Y

Sistemas de Informação

Enquadramento no ACM CCS 2012

The ACM Computing Classification System (CCS) Switch to Flat View Generate CCS Codes

General and reference	Hardware	Computer systems organization	Networks
Software and its engineering	Theory of computation	Mathematics of computing	Information systems
Security and privacy	Human-centered computing	Computing methodologies	Applied computing
Social and professional topics	What is the CCS? https://dl.acm.org/ccs/ccs.cfm		

Ênfases principais das UCs à responsabilidade da Área de Sistemas de Informação

Sistemas de Informação

Enquadramento no ACM CCS 2012

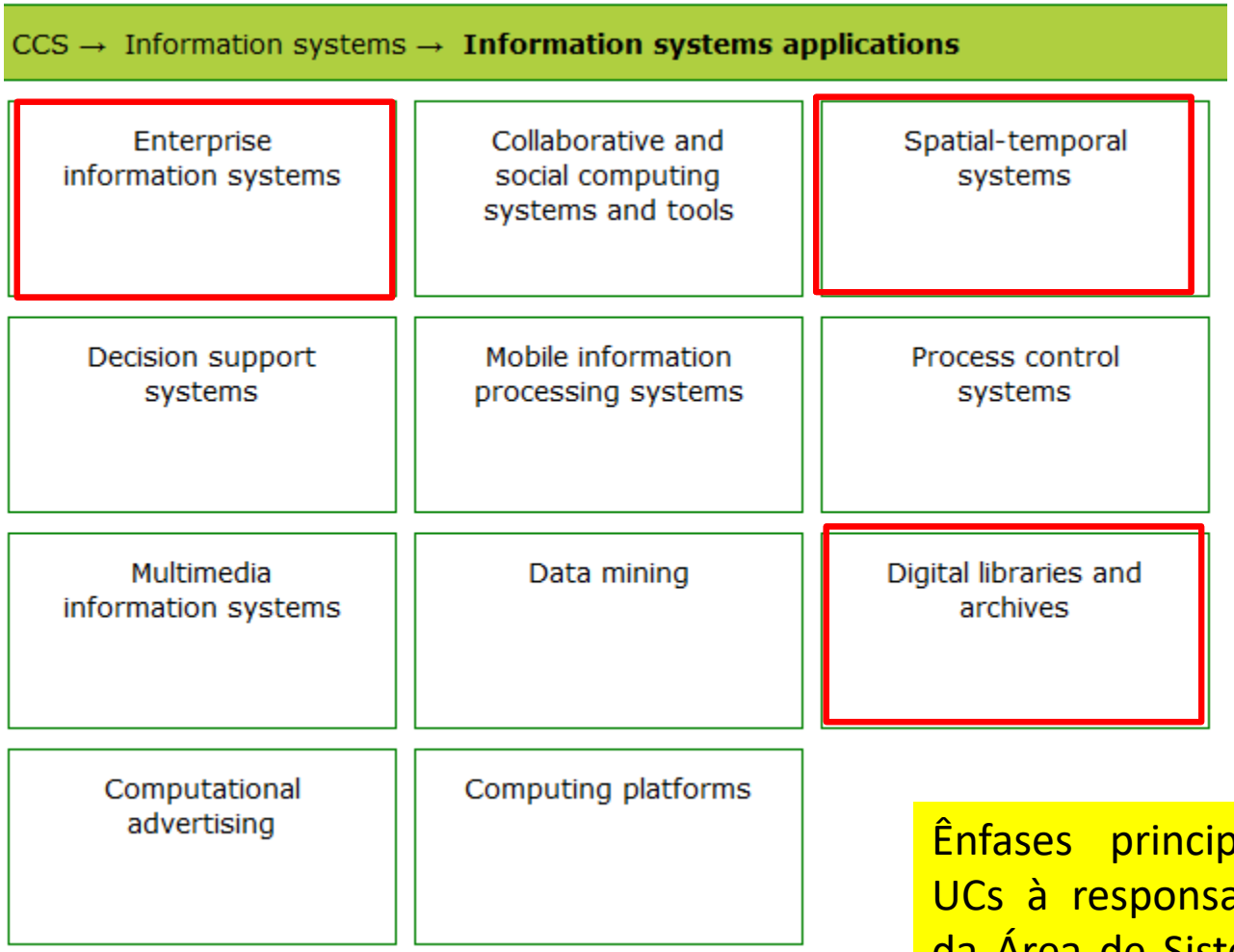
CCS → **Information systems**

Data management systems	Information storage systems	Information systems applications
World Wide Web	Information retrieval	

Ênfases principais das UCs à responsabilidade da Área de Sistemas de Informação

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Enquadramento no ACM CCS 2012



Ênfases principais das UCs à responsabilidade da Área de Sistemas de Informação

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Enquadramento no ACM CCS 2012

CCS → **Applied computing**		
Electronic commerce	Enterprise computing	Physical sciences and engineering
Life and medical sciences	Law, social and behavioral sciences	Computer forensics
Arts and humanities	Computers in other domains	Operations research
Education	Document management and text processing	

Ênfases principais das UCs à responsabilidade da Área de Sistemas de Informação

Sistemas de Informação

Alinhamento do MEIC com o MSIS 2016

https://msis2016review.wordpress.com/

"MSIS 2016: Global Competency Model for Graduate Degree Programs in Information Systems", is the latest in the series of reports that provides guidance for degree programs in the Information Systems (IS) academic discipline. The first of these reports (Ashenhurst, 1972) was published in the early 1970s, and the work has continued ever since both at the undergraduate and master's levels. The Association for Computing Machinery (ACM) has sponsored the reports from the beginning. Since the Association for Information Systems (AIS) was established in the mid-1990s, the two organizations have collaborated on the production of curriculum recommendations for the IS discipline. At the undergraduate level, both the Association for Information Technology Professionals (AITP) (formerly DPMA) and the International Federation for Information Processing (IFIP) have also made significant contributions to the curriculum recommendations.

MSIS 2016 is the seventh collaborative effort between ACM and AIS (following IS'97, IS 2002, and IS 2010 at the undergraduate level; MSIS 2000 and MSIS 2006 at the graduate level; and CC 2005 as an integrative document). Both ACM and AIS are global organizations that work to advance computing and its transformative uses. ACM's membership includes industry professionals, academics, and students worldwide, and it works in a broad spectrum of areas in computing. AIS is the premier global society for faculty members affiliated with IS, and it also serves students through a student chapter structure. The organizations complement each other's strengths and have been strong partners in educational initiatives since the 1990s. "

Sistemas de Informação

Alinhamento do MEIC com o MSIS 2016

MSIS 2016 (Master of Science in Information Systems) provides a competency model and curriculum guidance for master's level degree programs in information systems (IS). This joint effort by AIS and ACM builds on the foundation of four earlier graduate IS curriculum recommendations (…). MSIS 2016 does, however, break new ground in several important ways—most importantly by focusing on graduate competencies and competency areas and categories as its basic architectural building blocks and by offering a recommendation that has been specifically designed for the global IS community by a task force with broad cultural and geographic diversity. In addition, MSIS 2016 explicitly recognizes that business is not the only domain of practice for IS programs and considers alternatives such as healthcare, government, education, and law.

MSIS 2016 is based on a set of core premises, the most important of which include:

1. MSIS is a professional practice master's degree that always integrates the development of competencies in the realms of information systems (**including both computing and IT and IS management**), a specific domain of practice, and individual foundational competencies.

2. MSIS is based on a completed undergraduate degree that provides a foundation in all three major competency realms (see #1 above). Missing competencies can be developed with pre-program bridge courses.

3. MSIS does not have any general expectations regarding prior professional experience (although an individual program can set its own professional experience requirements).

4. The central element of this recommendation comprises specifications for a hierarchy of competency areas, competency categories, and sample competencies for IS. In addition, it provides general descriptions of required areas of individual foundational competencies and examples of areas of domain of practice competencies.

5. The target professional profiles of various MSIS programs vary (sometimes significantly).

6. Different professional profiles require different sets of competencies. MSIS 2016 specifies four levels at which a student can attain competencies in a category: Awareness, Novice, Supporting (role), and Independent (contributor). A competency profile specifies for each competency category the level the graduates of a program should attain.

7. A program can demonstrate compatibility with MSIS 2016 by showing that all of its graduates attain at least the minimum level of competency specified in this recommendation in each of the categories.

8. Competency categories and their attainment levels (see #6 above) form the foundation for determining modules (courses and equivalents) and their learning objectives. The structure, delivery modes, pedagogy, and content of the modules will.

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Alinhamento do MEIC com o MSIS 2016

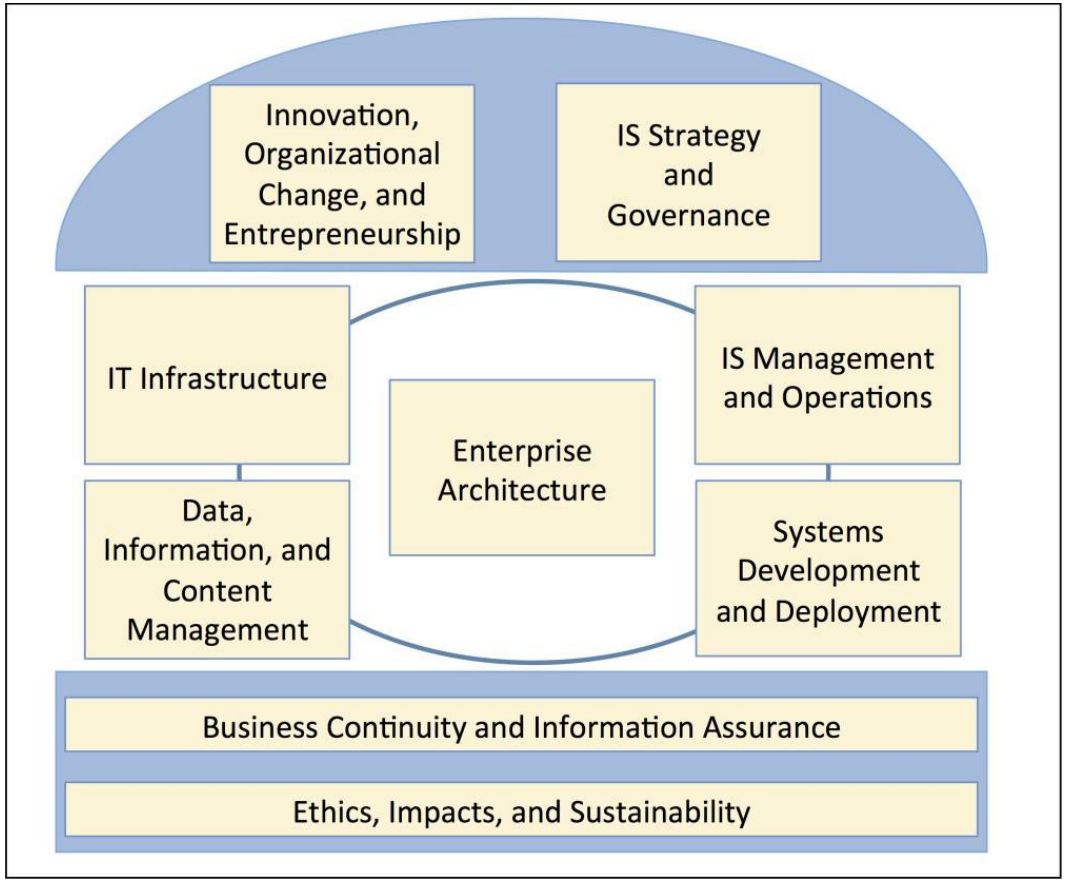
MSIS 2016 comprises nine IS **competency areas** (the full names of all are preceded by the phrase "competencies in," but we are not repeating it for the sake of brevity):

1. Business Continuity and Information Assurance
2. Data, Information, and Content Management
3. Enterprise Architecture
4. Ethics, Impacts, and Sustainability
5. Innovation, Organizational Change, and Entrepreneurship
6. IS Management and Operations
7. IS Strategy and Governance
8. IT Infrastructure, and
9. Systems Development and Deployment.

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Alinhamento do MEIC com o MSIS 2016



Global Competency Model for Graduate Degree Programs in Information Systems
The Joint ACM/AIS MSIS 2016 Task Force

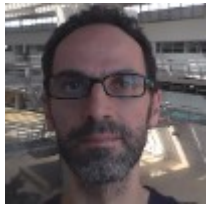
Sub-Áreas de Sistemas de Informação

Sub-áreas:

- Tecnologias de Sistemas de Informação
- Arquitectura e Gestão de Sistemas de Informação

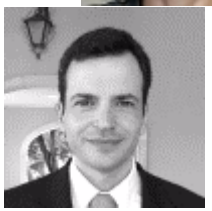
- Link no Fenix para o site da área cientifica: https://fenix.tecnico.ulisboa.pt/departamentos/dei/area-cientifica-de-sistemas-de-informacao

Quem Somos

















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Principais Unidades Curriculares de Sistemas de Informação

LEIC:

- BD - Bases de Dados
- AMS - Análise e Modelação de Sistemas

MEIC (e oferecidas ainda a outros Mestrados):

Transversal, genérica

- GPI - Gestão de Projectos Informáticos

Focadas em Tecnologia

- ADSI - Administração de Dados e Sistemas de Informação
- AID - Análise e Integração de Dados
- PRI - Processamento e Recuperação de Informação
- IE - Integração Empresarial

Focadas em Métodos e Técnicas

- FSI - Fundamentos de Sistemas de Informação
- AE - Arquitetura Empresarial
- OGFI - Organização e Gestão da Função Informática
- ETPN - Engenharia e Tecnologia de Processos de Negócio

Transversal, especializada

- TIS – Tecnologias de Informação em Saúde
- …

outras mais, especializadas…

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BD – Objectivos de Aprendizagem

1. Compreender os conceitos básicos necessários no **desenho e concepção de um sistema de gestão de base de dados.**
2. Dominar as questões práticas da sua **realização no âmbito de um projeto em equipa**.
3. Saber organizar bases de dados segundo o **modelo relacional**, cobrindo o desenho lógico das bases de dados (desenho do esquema) e sua implementação, bem como os sistemas de **gestão de transações**.
4. adquirir o conhecimento básico sobre sistemas de bases de dados paralelas e distribuídas, gestão de informação não estruturada e semiestruturada.

AMS – Objectivos de Aprendizagem

1. Saber desenvolver, integrado em equipas ou de forma autónoma, processos de **análise de requisitos** de **problemas de engenharia.**
2. Adquirir capacidade para desenvolver **modelos conceptuais de sistemas** candidatos a soluções para os problemas identificados.
3. Compreender a **metodologia de desenvolvimento de sistemas lógicos** (sistemas de software), ou sistemas com componentes físicos determinantes interagindo com sistemas lógicos, com ênfase para a análise de requisitos funcionais e não funcionais e desenho de modelos de estruturas e de comportamento;
4. Aplicar a metodologia de desenvolvimento de **sistemas de informação empresariais** (problemas em contexto organizacional, enquanto sistemas de processos de negócio cuja execução é feita com suporte a sistemas lógicos).

GPI – Objectivos de Aprendizagem

1. Conhecer os conceitos fundamentais da disciplina de gestão de projectos, nomeadamente ao nível das suas três variáveis de controlo: qualidade/requisitos, custo e tempo.
2. Conhecer as capacidades e conhecimentos específicos de um gestor de projectos informáticos.
3. Contextualizar a disciplina de gestão de projectos nos processos de desenvolvimento de software.
4. Conhecer e aplicar técnicas de planeamento e estimação de projectos de informáticos.
5. Conhecer os paradigmas emergentes da gestão de projectos.
6. Conhecer e usar ferramentas de gestão de projectos e de trabalho colaborativo.

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ADSI – Objectivos de Aprendizagem

1. compreender **os mecanismos internos de um SGBD** relacional, nomeadamente a gestão de armazenamento, a indexação, o processamento e optimização de interrogações, a gestão de transacções, do controlo de concorrência e gestão de recuperação;
2. compreender as tarefas envolvidas na **administração de bases de dados**;
3. **optimizar o acesso à informação** em bases de dados que armazenam grandes quantidades de dados; e
4. adquirir o conhecimento básico sobre as várias **arquitecturas de bases de dados paralelas e distribuídas**, convencionais (SQL) e não convencionais (NoSQL).

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AID: Objectivos de Aprendizagem

1. Entender como **integrar dados** provenientes **de fontes heterogéneas**;
2. Conhecer os **paradigmas de integração de dados**;
3. Pré-processar os dados para **carregamento num armazém de dados**, resolvendo os erros e inconsistências encontrados e eliminando elementos duplicados;
4. **Analisar os dados** disponíveis num armazém de dados.

PRI: Objectivos de Aprendizagem

1. Projetar soluções para o processamento, gestão e interrogação de grandes volumes de **informação não estruturada ou semi-estruturada**;
2. **Classificar e agrupar automaticamente** conjuntos de recursos (e.g., grandes conjuntos de documentos de texto) através de características descritivas;
3. Conceber sistemas para a **recuperação e filtragem da informação** relevante existem em grandes colecções, com base em termos chave, com base em exemplos, ou com base em perfís dos utilizadores;
4. Conceber sistemas para a **extracção de informação** de documentos de texto, ou da Web;
5. Avaliar comparativamente diferentes **sistemas para a extração, filtragem e recuperação de informação** relevante.

IE - Objectivos de Aprendizagem

1. Adquirir uma visão abrangente dos principais conceitos e soluções tecnológicas existentes na área de integração;
2. Desenvolver uma **visão sistemática e orientada aos processos** sobre a forma de abordar problemas de integração;
3. Adquirir um conhecimento prático sobre as **plataformas de integração actuais** com base em guias e projecto laboratoriais;
4. Compreender o papel crucial que as soluções de integração têm no desenho e **implementação de processos de negócio** numa organização

FSI – Objectivos de Aprendizagem

1. Fornecer uma **abordagem sociotécnica aos sistemas de informação organizacionais**; entender como e porquê os sistemas de informação são hoje usados nas organizações.
2. Explicar a relações entre os componentes de tecnologia, de pessoas e organizacionais dos sistemas de informação empresariais.
3. Conhecer os principais componentes de tecnologia dos sistemas de informação
4. Entender como as empresas utilizam os sistemas de informação para suportar as suas atividades e criar vantagens competitivas.
5. Entender como os sistemas de informação permitem criar novas formas de comércio entre indivíduos, organizações e governos.
6. Conhecer as novas tecnologias que permitem novas formas de comunicação, colaboração e parcerias.
7. Entender como os sistemas de informação permitem criar relações com clientes e fornecedores, e como são utilizados para reforçar as estruturas organizacionais e os processos de negócio
8. Entender como os sistemas de informação podem suportar a tomada de decisão nos diferentes níveis e funções das organizações
9. Compreender como as organizações desenvolvem e adquirem tecnologias e sistemas de informação.
10. Compreender o valor dos investimentos em sistemas de informação, bem como aprender a preparar um plano de negócios para um novo sistema de informação, incluindo a estimativa de custos e benefícios.
11. Mitigar os riscos, bem como planear e recuperar de desastres.
12. Entender como garantir a segurança dos sistemas de informação, tendo em conta tanto aspetos tecnológicos como humanos.
13. Avaliar as questões éticas dos sistemas de informação, e o impacto dos sistemas de informação na fraude e no crime.

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AE – Objectivos de Aprendizagem

1. Compreender o ciclo de vida da engenharia empresarial, incluindo a governação e os processos de transformação organizacionais.
2. Compreender as teorias fundamentais subjacentes engenharia empresarial.
3. Compreender e aplicar os princípios de arquitetura empresarial.
4. Compreender e usar as linguagens de modelação de arquitetura empresarial, nomeadamente o ArchiMate e o DEMO.
5. Compreender e utilizar modelos e molduras de referência das organizações por indústria.
6. Compreender e aplicar as técnicas de alinhamento empresarial.
7. Analisar casos de estudos de organizações reais.

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OGFI – Objectivos de Aprendizagem

1. Compreender as diversas funções e atividades na área de sistemas de informação (SI) incluindo o papel da gestão da informática e do director de SI, a estrutura da gestão dos SI dentro de uma organização, e a gestão dos profissionais de SI.
2. Ver uma organização pelo ponto de vista da administração de topo para decidir como os SI devem suportar os processos de negócio, quer os principais, quer os de suporte, quer os de gestão de clientes e fornecedores.
3. Compreender os conceitos de economia da informação ao nível da organização.
4. Perceber como os SI representam uma importante fonte de vantagem competitiva para as organizações.
5. Estruturar as actividades relacionadas com os SI para maximizar o valor dos SI dentro e fora das organizações.
6. Compreender as tecnologias de informação existentes e emergentes, as funções dos SI, e o seu impacto sobre as operações da organização.
7. Avaliar as questões e os desafios associados com a incorporação dos SI numa organização.
8. Entender como as decisões estratégicas são feitas sobre a aquisição de SI, incluindo a capacidade de avaliar as diferentes opções de fornecimento.
9. Aplicar informação às necessidades de diferentes indústrias e áreas.
10. Compreender o papel do controlo e as frameworks de gestão de serviços de informática pela perspectiva da gestão dos SI numa organização.

ETPN – Objectivos de Aprendizagem

1. Compreender o papel dos processos de negócio numa organização.
2. Compreender a relação dos processos de negócio com a arquitectura empresarial e as tecnologias de informação de suporte.
3. Analisar e desenhar processos de negócio através de linguagens de modelação adequadas.
4. Analisar processos de negócio através de técnicas manuais, semi-automáticas e automáticas para análise de informação, incluindo técnicas de princípios de arquitectura e de mineração de processos.
5. Redesenhar e optimizar processos de negócio mantendo a sua rastreabilidade para os requisitos de transformação.
6. Compreender o papel da arquitectura dos sistemas de gestão de processos de negócio (BPMS).
7. Compreender o papel prático das ferramentas de BPM, incluindo as ferramentas de modelação e análise de processos.

TIS: Objectivos de Aprendizagem

1. Dotar os alunos com os princípios e conceitos fundamentais relativos à utilização das tecnologias de informação em saúde.
2. Familiarização com as potencialidades do uso das tecnologias de informação na investigação em biomedicina
3. Compreender o papel crucial que as TIC hoje representam nas várias vertentes da prestação de cuidados de saúde.

Saídas Profissionais em Sistemas de Informação

- Administração de Sistemas de Informação e Bases de Dados
- Concepção de sistemas
  - de gestão de processos e de gestão de conhecimento
  - de informação e apoio à decisão
- Especialistas em TIC para sectores da economia
  - Saúde
  - Telecom
  - Energia
  - Transportes
  - …

Saídas Profissionais em Sistemas de Informação

- **Empregados** por:
  - **Empresas de consultoria**
    - Em sistemas de informação
      - Análise desenvolvimento e apoio à exploração de aplicações de negócio
    - Estratégica
    - **Grande procura de engenheiros do IST para projetos internacionais!!!**
  - **Departamentos de sistemas de informação**/organização de grandes empresas

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Saídas Profissionais em Sistemas de Informação

- Alguns criam também **as suas próprias empresas**!
- Mais cedo ou mais tarde, **muitos engenheiros ascendem a cargos de condução de negócios**.
  - As tecnologias de gestão das organizações, da sua informação e conhecimento tornar-se-ão competências críticas.

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Tecnologias de Sistemas de Informação
Modelo de Dados

- Um ***modelo de dados (data model)*** é uma colecção de conceitos para descrever dados.
- Um ***esquema (schema)*** é uma descrição de uma colecção específica de dados, usando um dado modelo de dados.

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Tecnologias de Sistemas de Informação
Modelo Semântico de Dados

- Um **modelo semântico de dados** é um modelo mais abstracto que:
  - Está mais perto da visão do utilizador
  - Tem mais primitivas
  - É um bom ponto de partida
  - No final é traduzido para um modelo de dados
- **Modelo Entidade-Associação (EA)**
  - *Entity-relationship (ER) model*
  - Descreve de uma forma gráfica as entidades e as relações entre elas

Tecnologias de Sistemas de Informação
Modelo Entidade-Associação

- Modela uma organização como coleção de entidades e relacionamentos (associações).
  - **Entidade**: "coisa" ou "objeto" na organização que é distinguível de outros objetos
    - Descrito através de um conjunto de atributos
  - **Associação**: conjunto de relacionamentos entre conjuntos de entidades similares

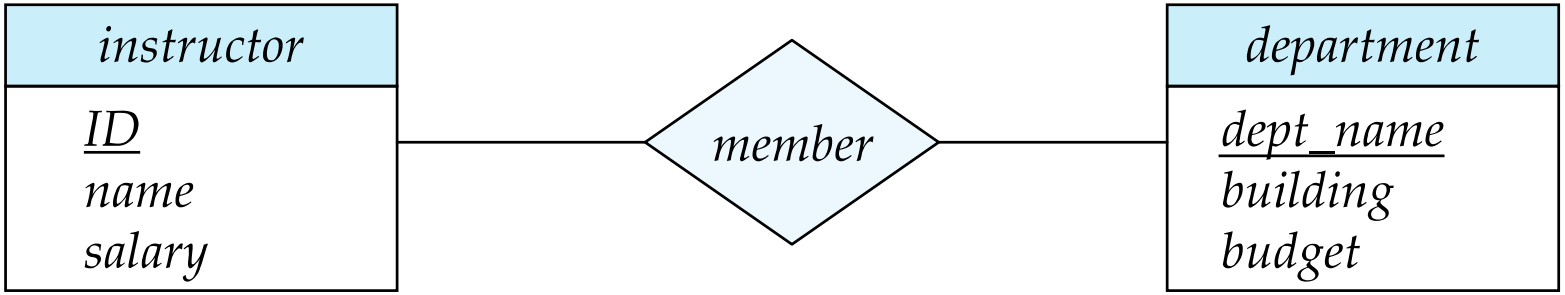


Diagrama entidade-associação



Tecnologias de Sistemas de Informação
Modelo Relacional

- Composto por **relações**
- O **esquema de uma relação** especifica:
  - O seu nome
  - O nome de cada atributo (ou campo)
  - O tipo de cada atributo
- Exemplo:

  Students(*sid: string, name: string, login: string, age: integer, gpa: real)*

  ** Age está incorrecto usado apenas para não complicar*

Exemplar de uma relação (tabela)

sid	*name*	*login*	*age*	*gpa*
53666	Jones	jones@cs	18	3.4
53688	Smith	smith@ee	18	3.2
53650	Smith	smith@math	19	3.8
53831	Madayan	madayan@music	11	1.8
53832	Guldu	guldu@music	12	2.0

- Cada linha na relação *Students* é um registo que descreve um aluno
- Cada linha segue o esquema da relação *Students*

Modelos de Dados

- Conjunto de conceitos e métodos para descrever
  - Dados
  - Relacionamentos
  - Semântica dos dados
  - Restrições

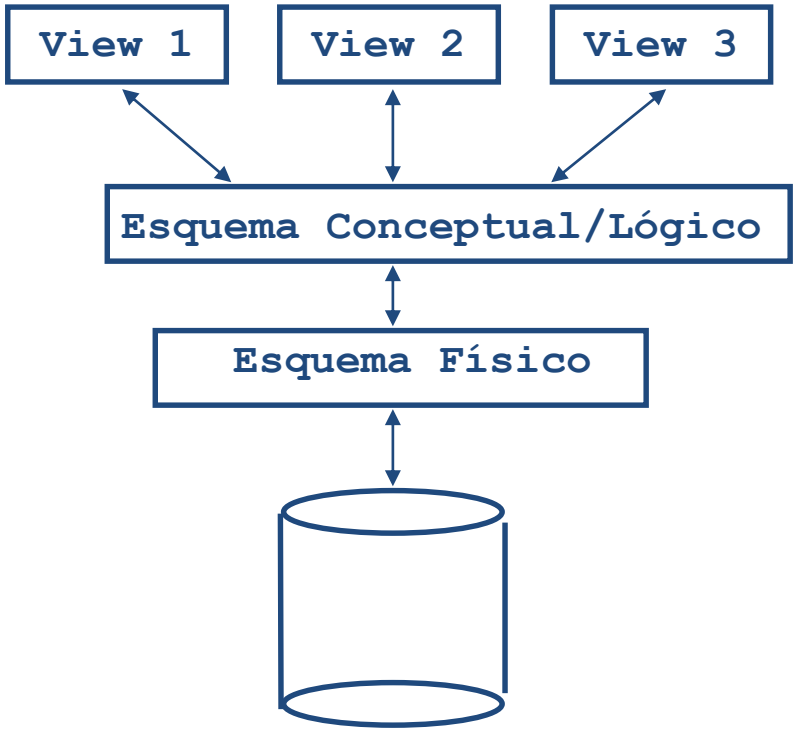
Modelos de Dados:

- Relacional
- Entidade-Relação
  (para concepção de bases de dados)
- Baseados em Objectos
  (Object-oriented e Object-relational)
- Semi-estruturados
  (XML, JSON,)
- Históricos:
  em Rede, hierárquico

Níveis de Abstracção
Modelo ANSI/SPARC

Muitas *views*, um só *esquema conceptual/lógico* e um só *esquema físico*.

- Esquema externo (Views) descreve como os utilizadores vêm os dados.
- Esquema conceptual projecta-se na estrutura lógica.
- Esquema Físico descreve ficheiros (tabelas e índices).



DDL: manipulação esquemas
DML: manipulação dados

Independência dos Dados

- A arquitectura em 3 níveis de esquemas (ou modelos) possibilita:
- **Independência dos dados lógicos**
  - é possível estender as estruturas de dados sem modificar os programas
- **Independência dos dados físicos**
  - é possível alterar a organização física dos dados sem modificar o esquema conceptual.



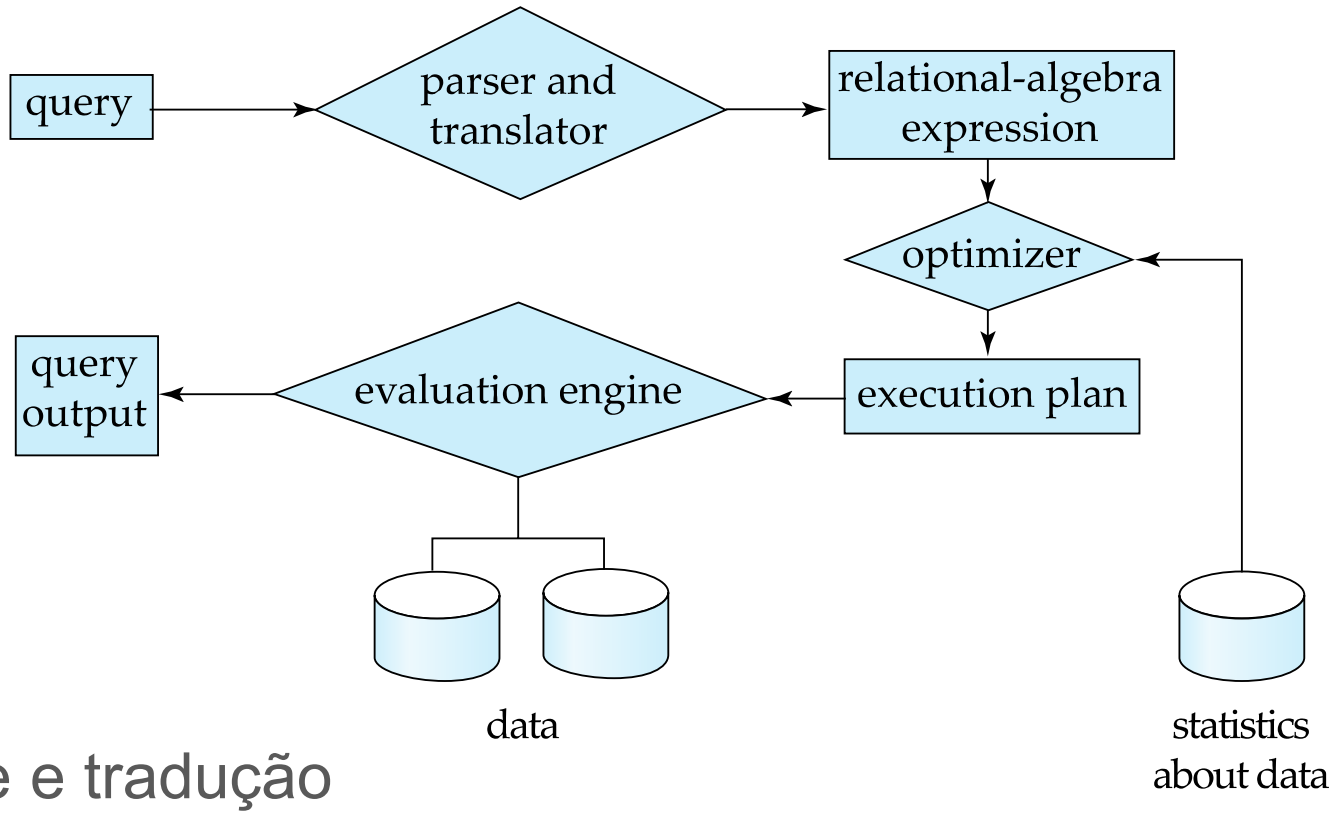
Interrogações ao SGBD

- Exemplo:

  Qual a fracção de estudantes na disciplina de IEI que obteve uma nota superior a 16?

- Tradução para a linguagem de interrogação do SGBD
  - Nos SGBD relacionais usa-se a linguagem **SQL (Structured Query Language)**
  - O SGBD tenta executar as interrogações da forma mais eficiente

Processamento de Interrogações



1. Análise e tradução
2. Otimização
3. Avaliação

Gestão de Transações

Uma **transação** é uma qualquer execução de uma aplicação no SGBD

Exemplos:

- Transferência de dinheiro
  - Com falha do sistema
- Compra de bilhete
  - Com acesso concorrente

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Tecnologias de Sistemas de Informação

Quem usa Bases de Dados ?

1. Construtores
2. Utilizadores (de aplicações)
3. **Programadores (de aplicações)**
   - Participam na concepção do modelo lógico do sistema de informação.
4. **DBA – Administradores de Bases de Dados**
   - Desenho lógico e físico
   - Segurança e autorização
   - Disponibilidade e recuperação em caso de falha
   - Afinação
   - DBA dedicado só existe nos grandes sistemas.

Dados Semi-Estruturados (XML)

```
<?XML version='1.0' ?>
<?xml:namespace name="urn:..../" as="s"/?>
<s:schema id='ExampleSchema'>

<elementType id="author">
   <string/>
</elementType>

<elementType id="Book">
   <element
        type="#author"
        occurs="ONEORMORE"/>
</elementType>

</s:schema>
```

```
<Book>

   <author>Henry Ford
    </author>
   <author>Samuel Crowther
    </author>

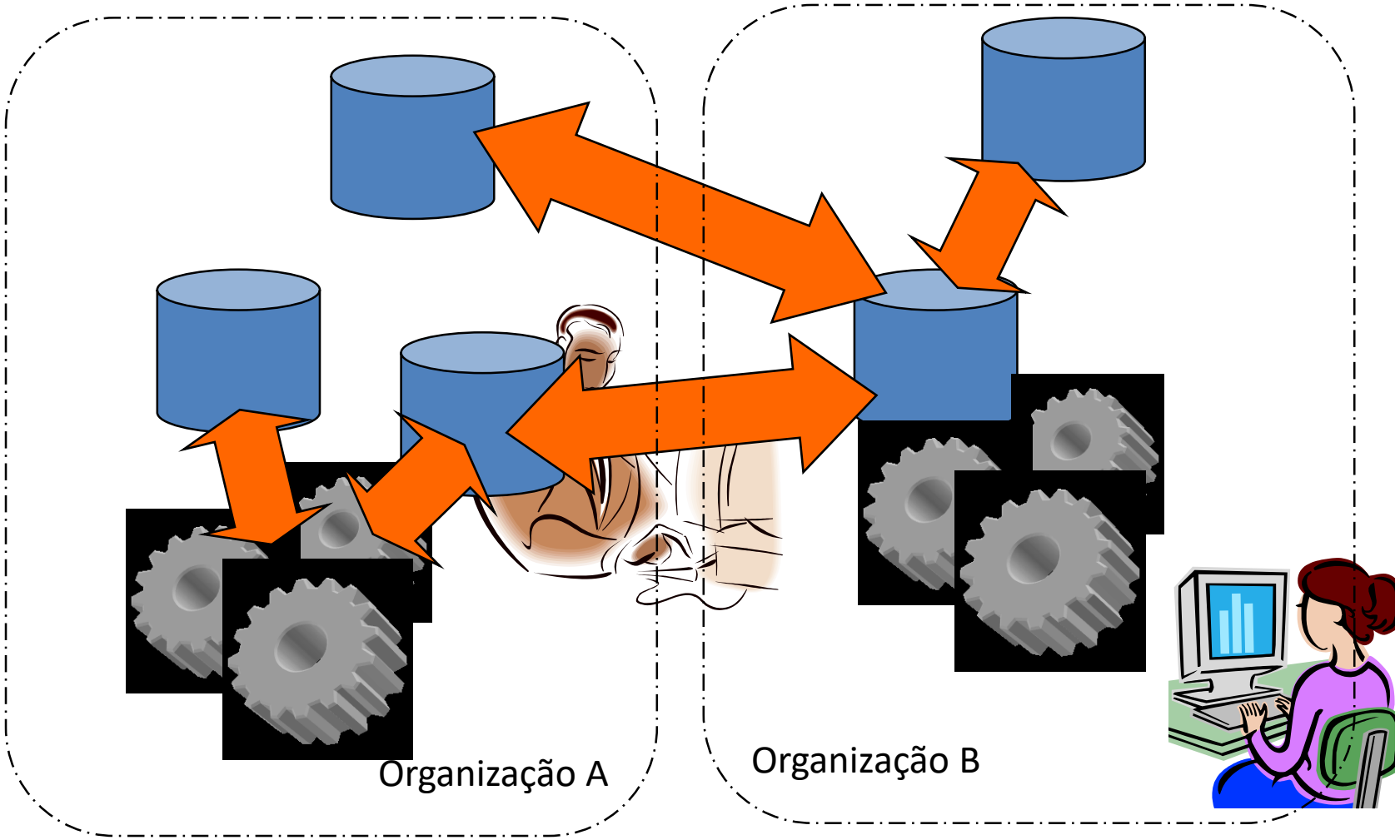
</Book>
```



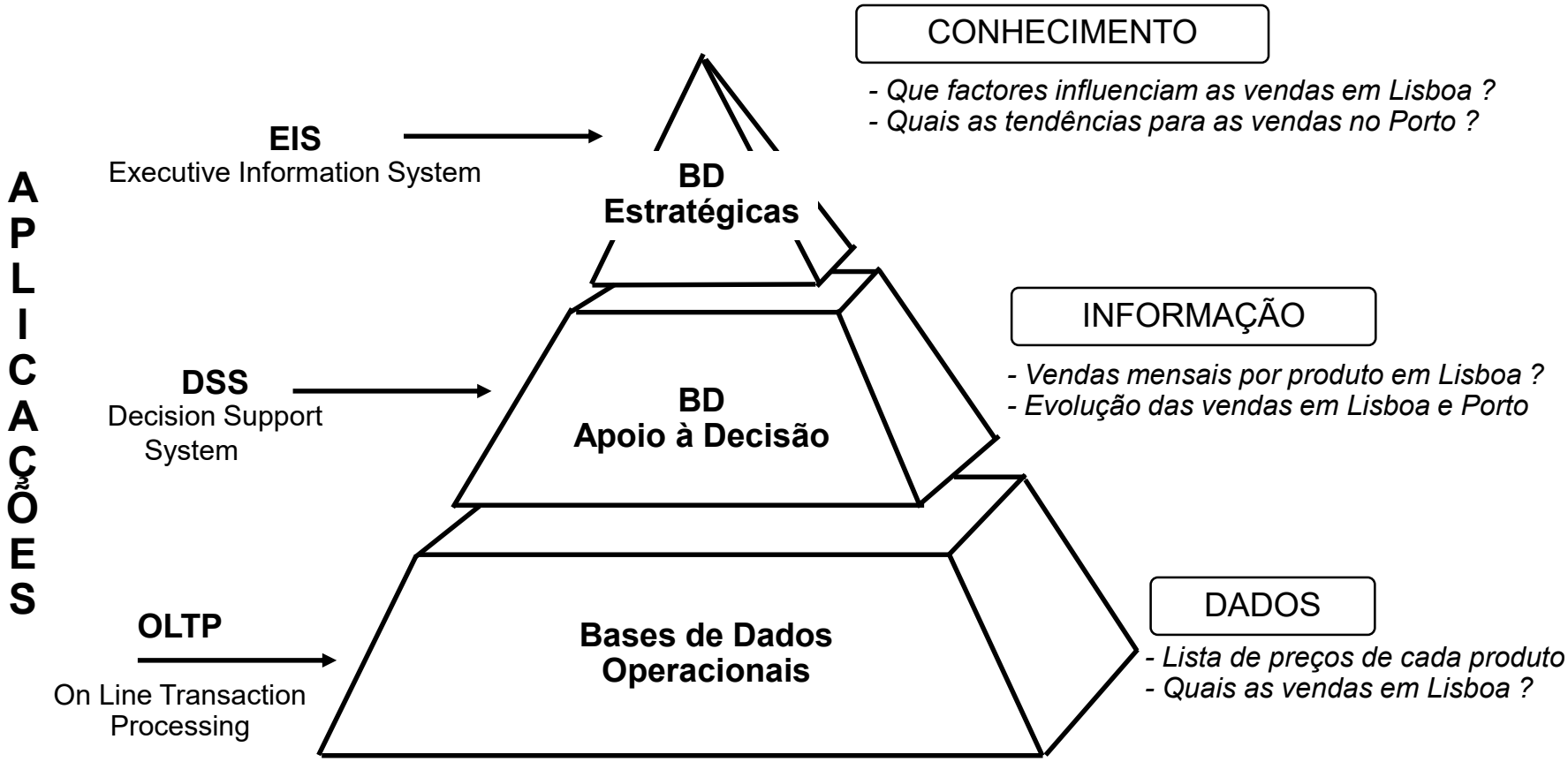
Dados Semi-estruturados

- Auto-descrição da informação (catálogo)
  - XML introduz mesma capacidade para dados semi-estruturados.
- Isolamento entre programas e dados (abstracção dos dados)
  - XML introduz mesma capacidade para dados semi-estruturados.
- Suporte de vistas
- Partilha de dados e processamento transaccional

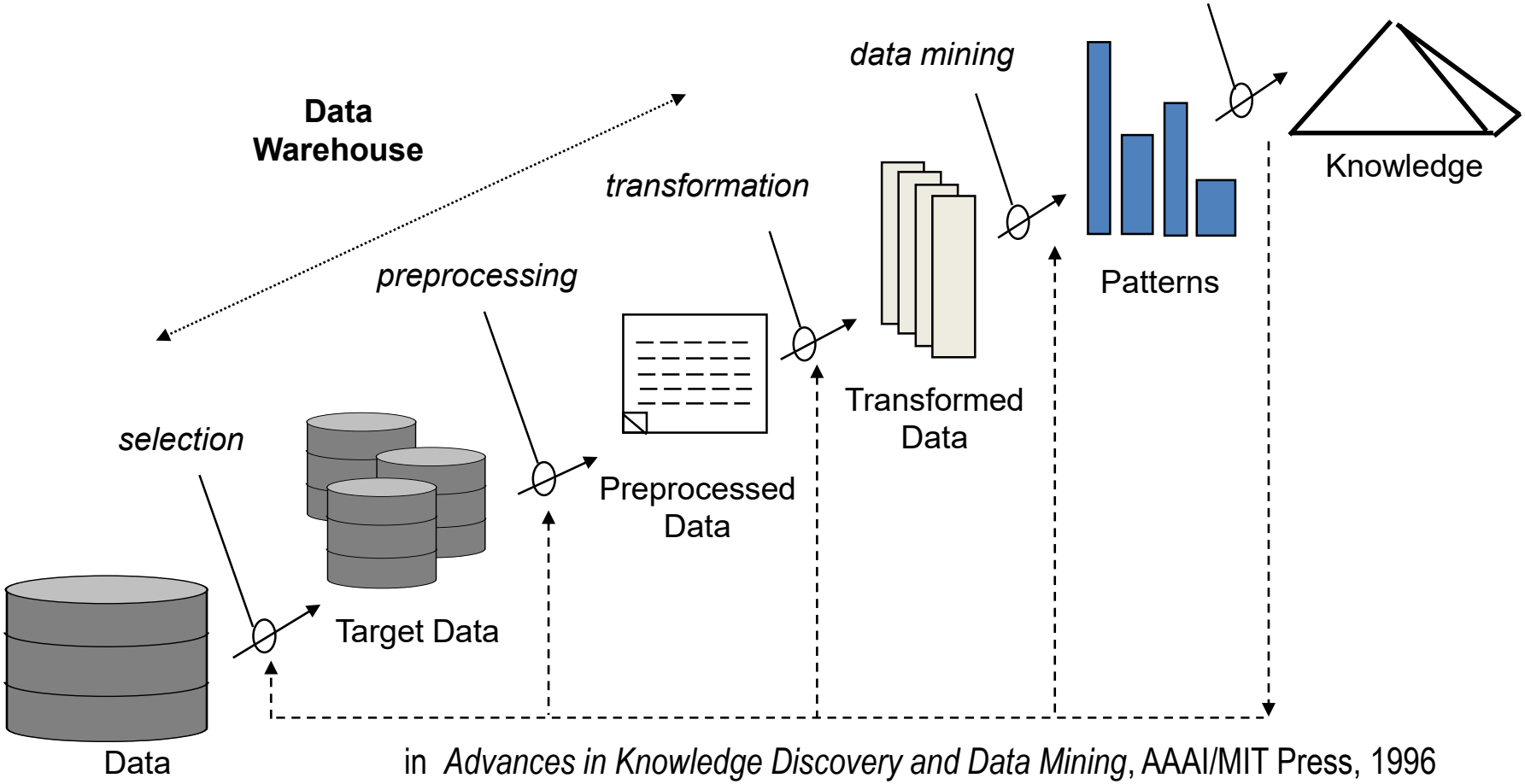
Tecnologia dos Sistemas de Informação
Os grandes desafios estão na INTEGRAÇÃO



Terminologia dos “níveis” de dados e sistemas de informação



Processo de descoberta de conhecimento



in *Advances in Knowledge Discovery and Data Mining*, AAAI/MIT Press, 1996



Concorrência e Robustez

- Como é que um SGBD (Sistema de Gestão de Bases de Dados) permite o acesso concorrente aos dados?
- Como é que este protege os dados num caso de uma falha do sistema?



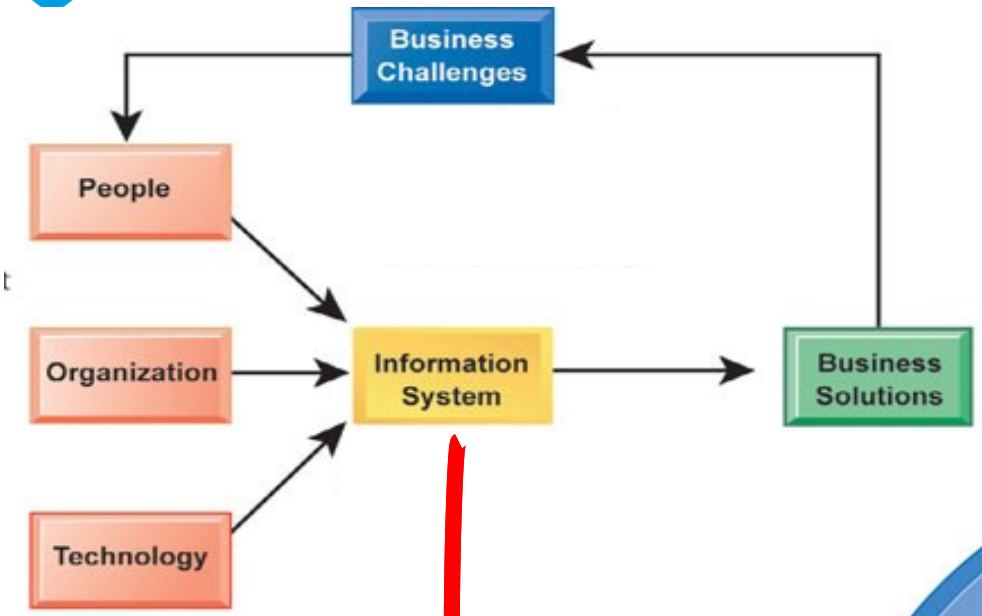
Eficiência e Escalabilidade

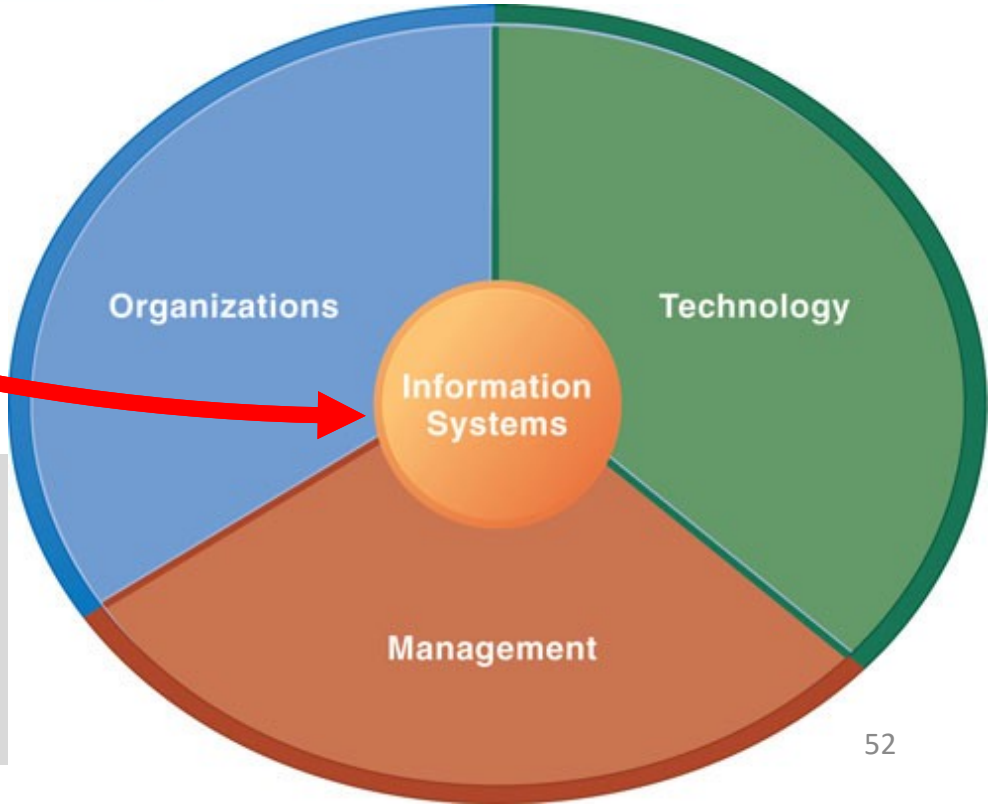
- Como é que um SGBD (Sistema de Gestão de Bases de Dados) armazena grandes colecções de dados e executa interrogações nesses dados de uma forma eficiente?

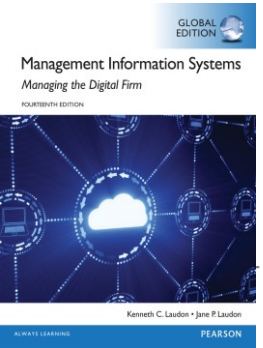
Porquê aprender Sistemas de Informação?

- Da computação à informação
  - aplicações “científicas” vs. a web
- Dados aumentam em diversidade e volume
  - Bibliotecas Digitais, vídeo interativo, genoma humano
  - o número de aplicações está em explosão!
- SGBD aplica conceitos que cobrem a quase totalidade das bases da Informática
  - Sistemas Operativos, Linguagens, Teoria, IA, Multimédia, Lógica

Arquitetura e Gestão de Sistemas de Informação



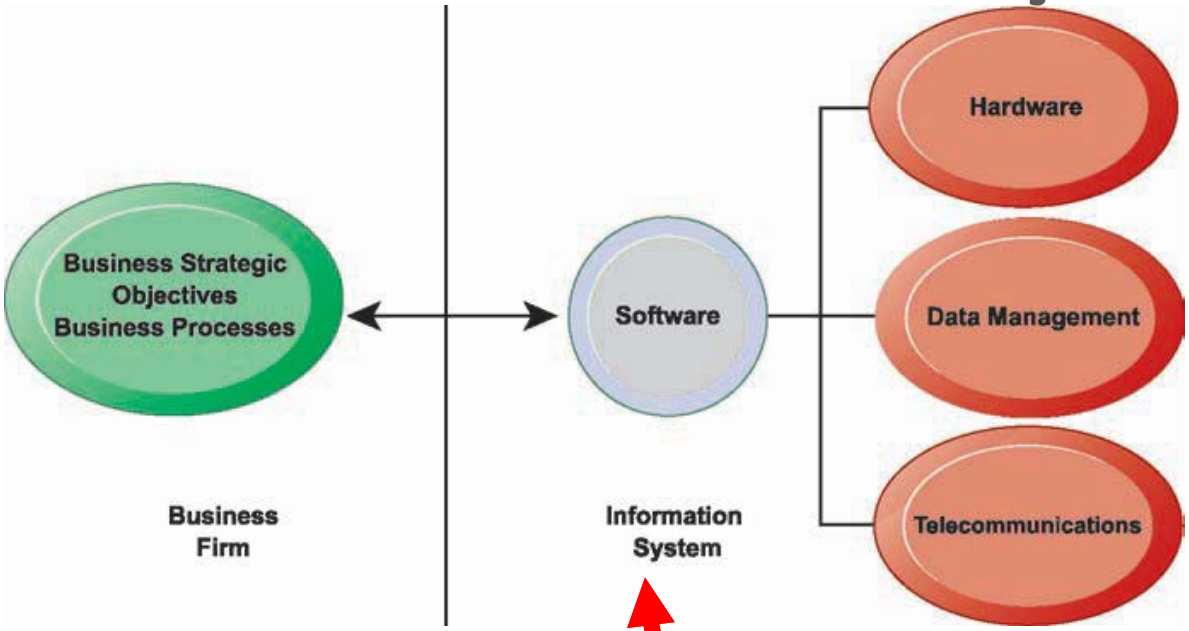


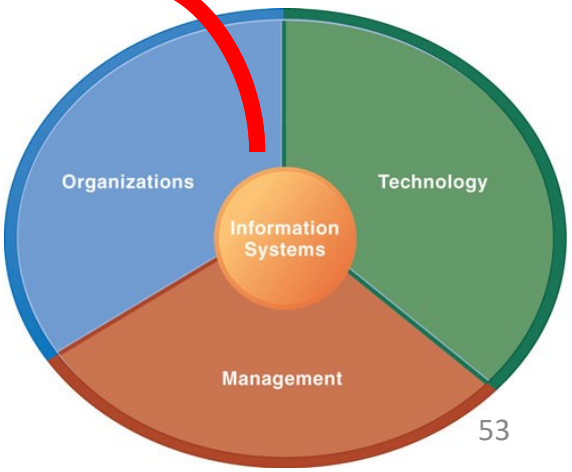


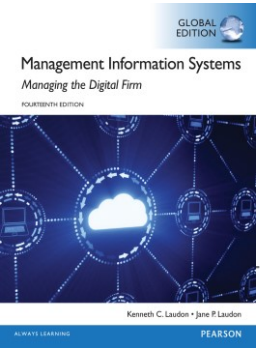
…management, organization, and technology elements work together to create an **information system solution to the business challenges**…

Arquitetura e Gestão de Sistemas de Informação

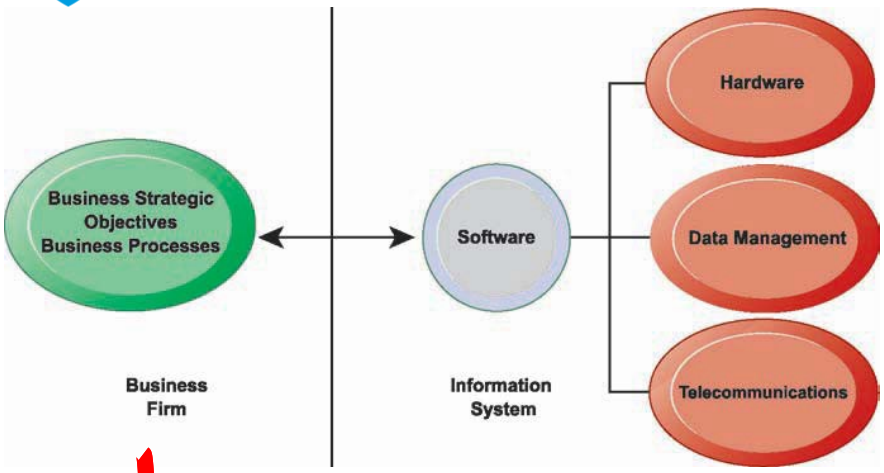
… there is a growing interdependence between a firm's information systems and its **business capabilities**. Changes in strategy, rules, and business processes increasingly require changes in hardware, software, databases, and telecommunications. **Often, what the organization would like to do depends on what its systems will permit it to do**.



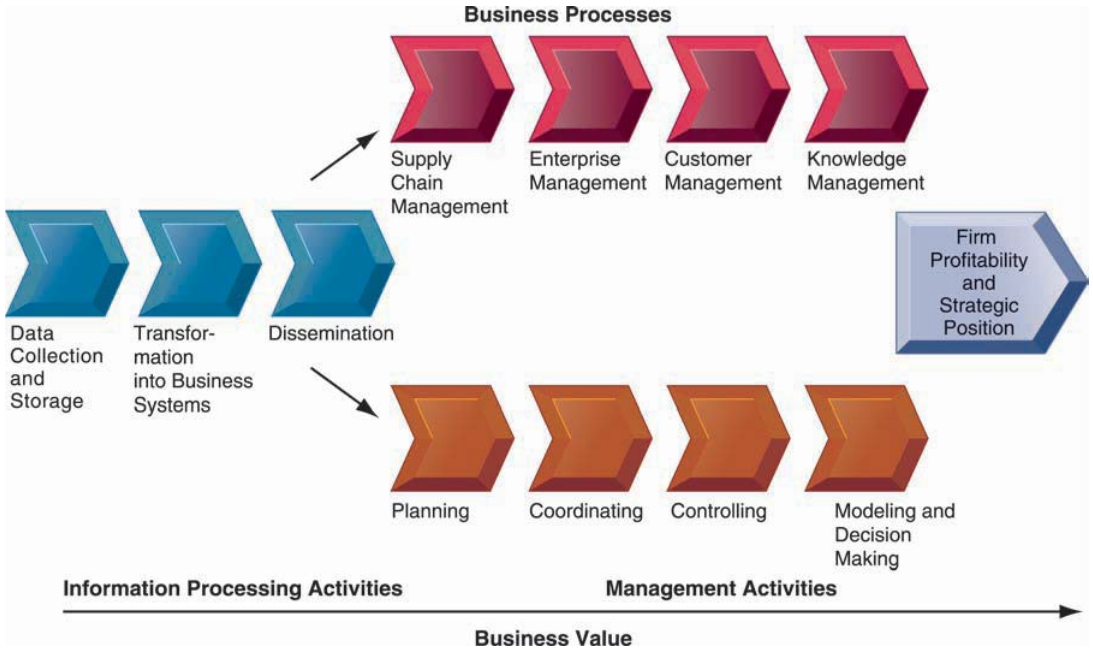


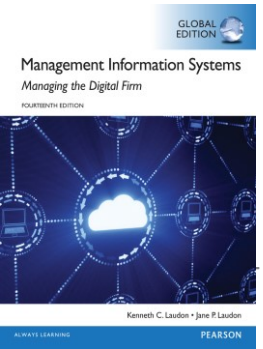


Arquitetura e Gestão de Sistemas de Informação

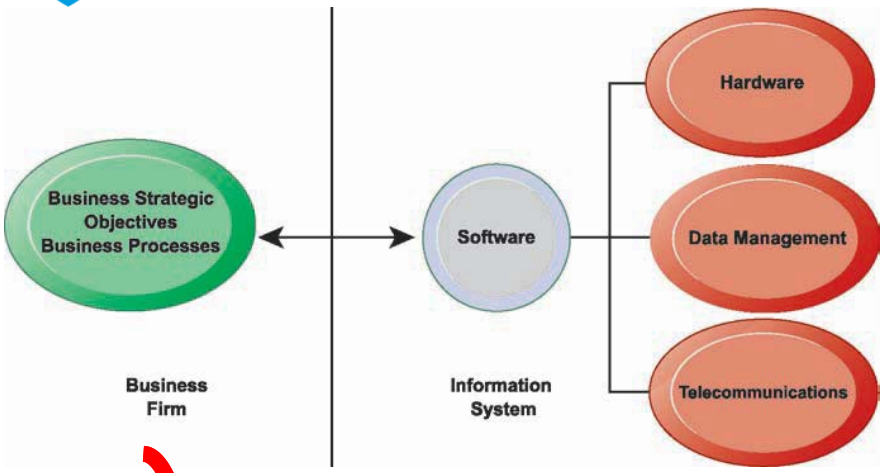


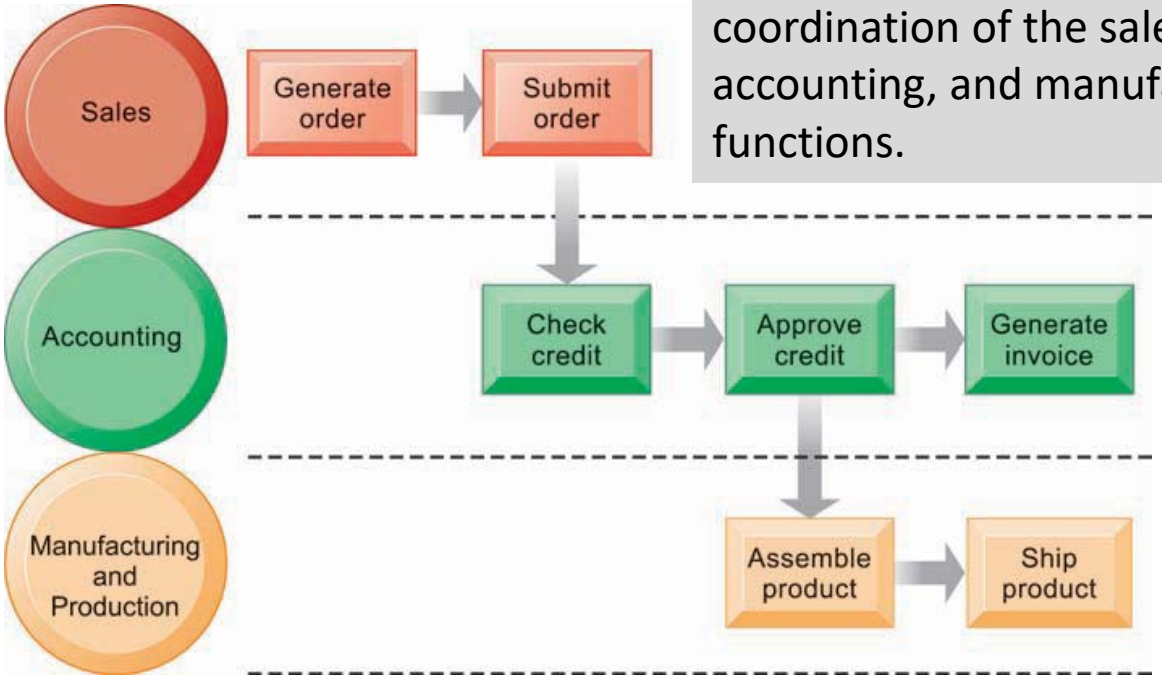
From a business perspective, information systems are part of a series of value-adding activities for acquiring, transforming, and distributing **information that managers can use** to improve decision making, enhance organizational performance, and, ultimately, increase firm profitability.



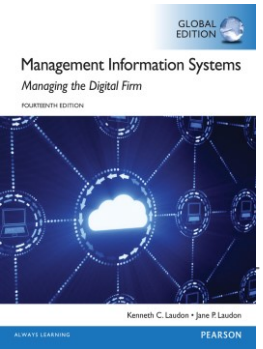


Arquitetura e Gestão de Sistemas de Informação

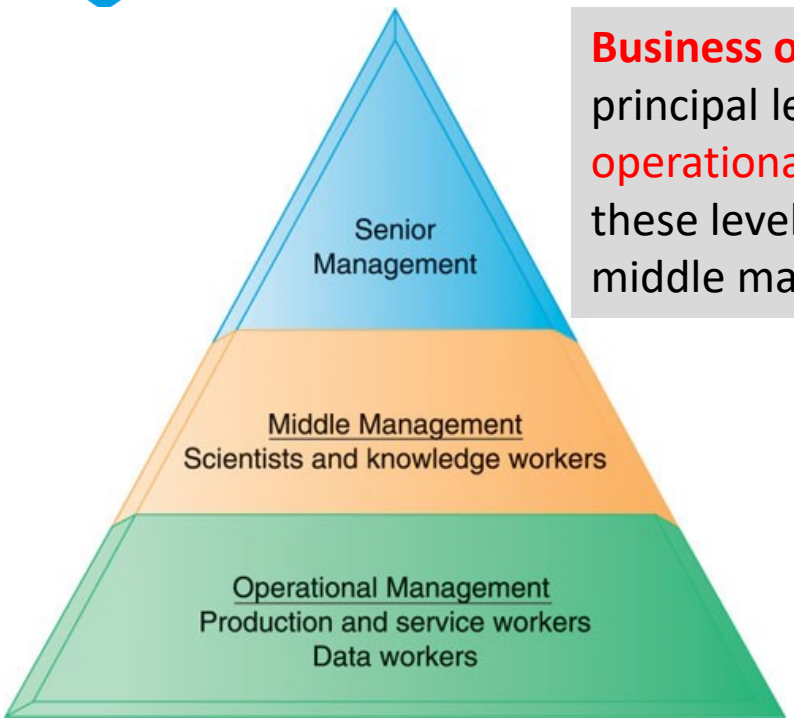




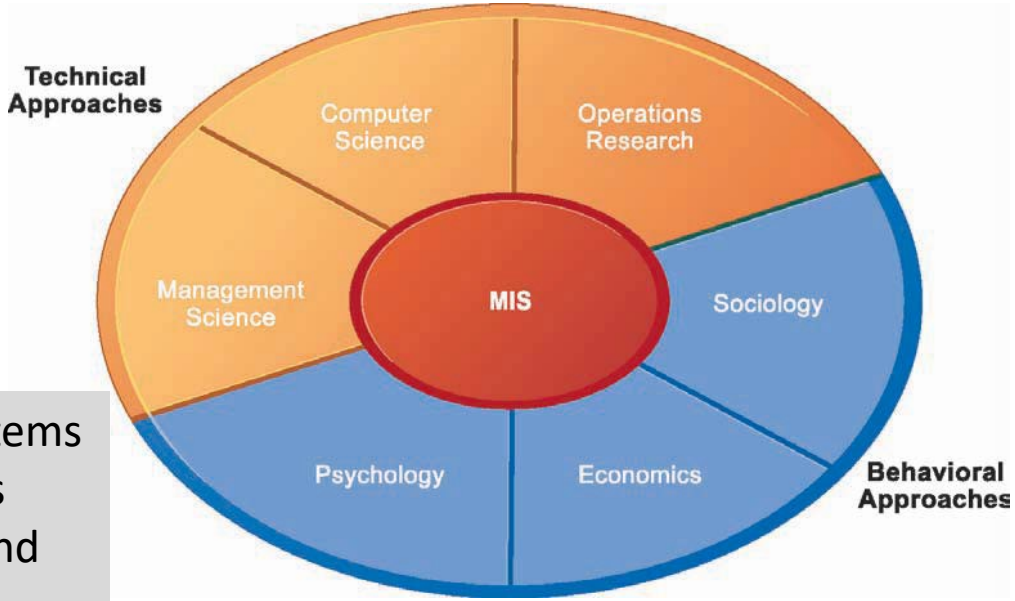
Business Processes:
Fulfilling a customer order involves a complex set of steps that requires the close coordination of the sales, accounting, and manufacturing functions.

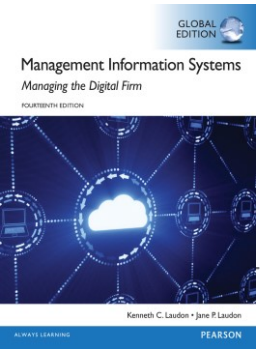


Arquitetura e Gestão de Sistemas de Informação



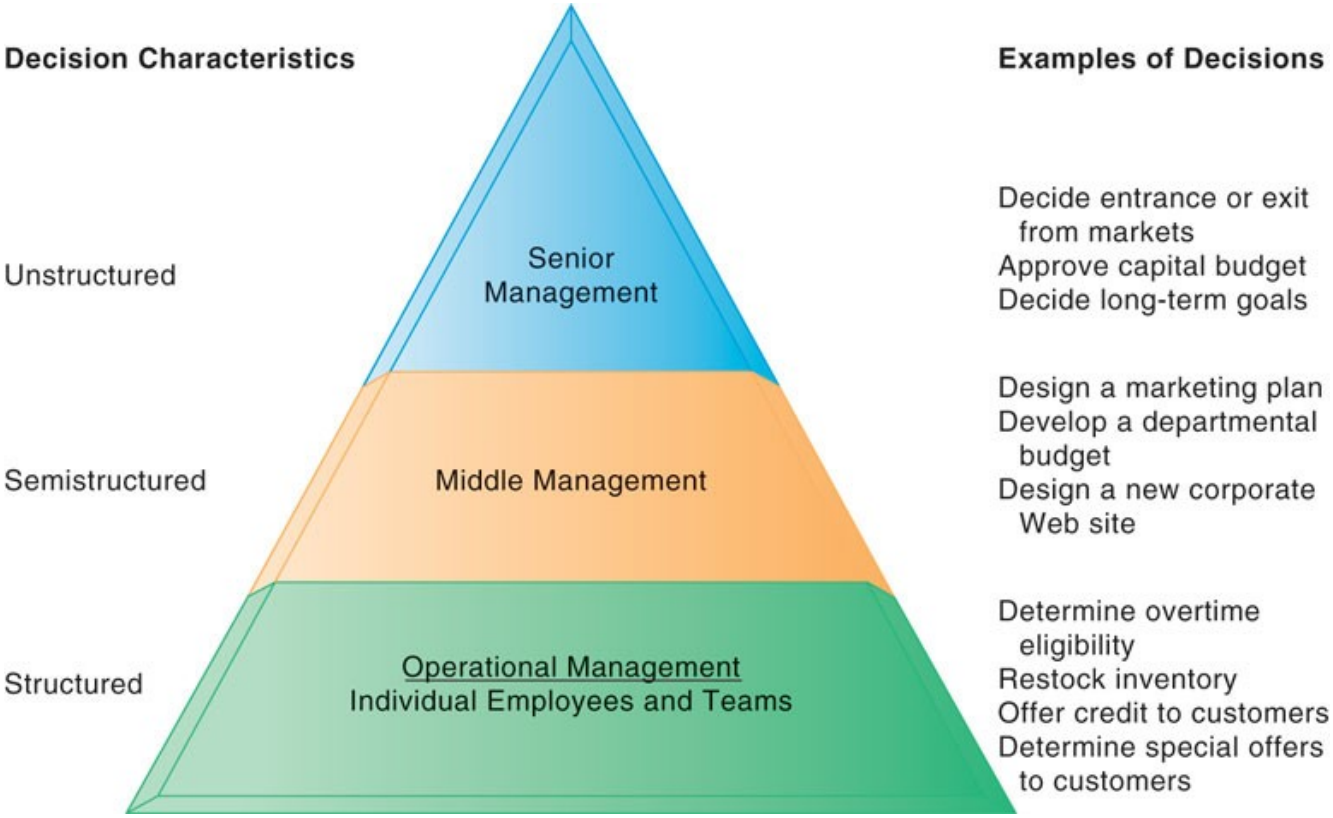
Business organizations are hierarchies consisting of three principal levels: senior management, middle management, and operational management. Information systems serve each of these levels. Scientists and knowledge workers often work with middle management.



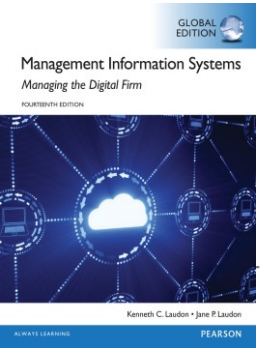


The study of information systems deals with issues and insights contributed from technical and behavioral disciplines.

Arquitetura e Gestão de Sistemas de Informação



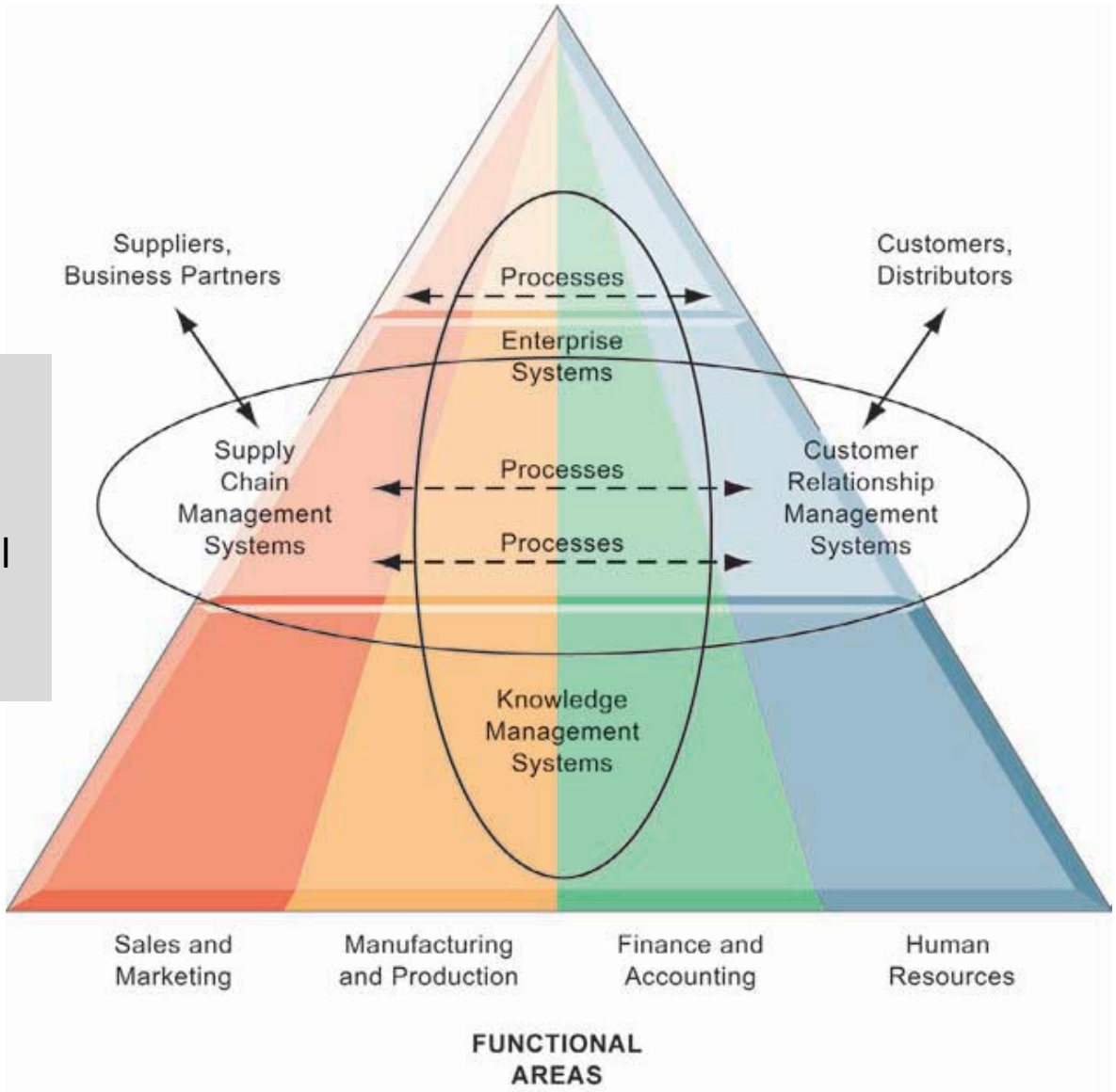
Sugestão: comparar com "slide" 47…

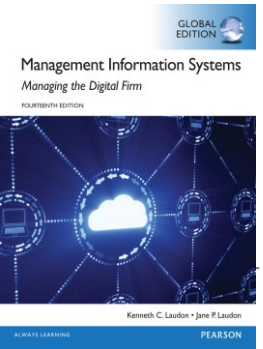


Senior managers, middle managers, operational managers, and employees have different types of **decisions and information requirements**

Arquitetura e Gestão de Sistemas de Informação

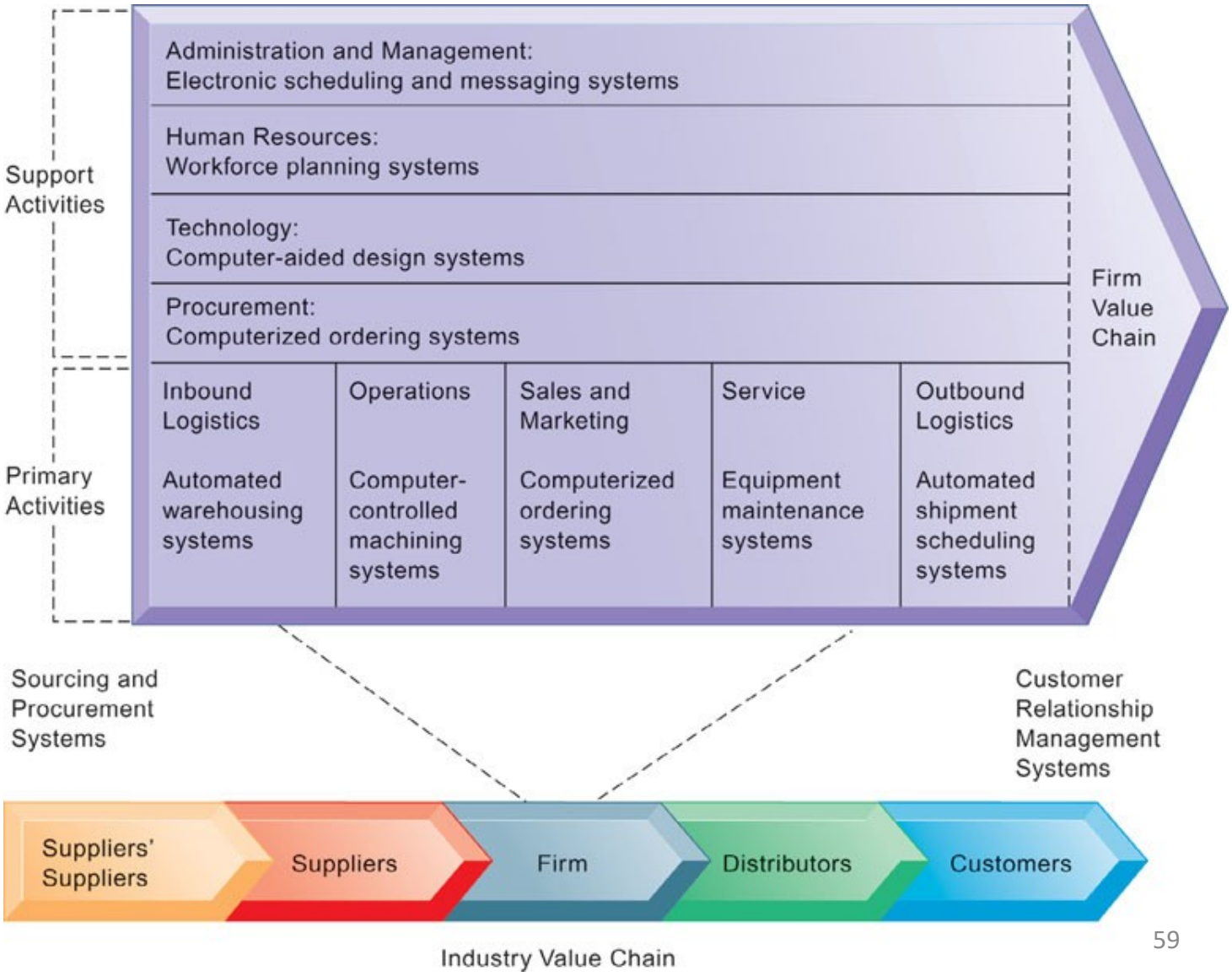
Enterprise applications automate processes that span multiple business functions and organizational levels **and may extend outside the organization**

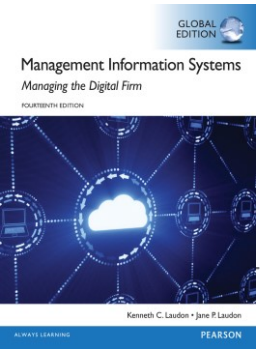




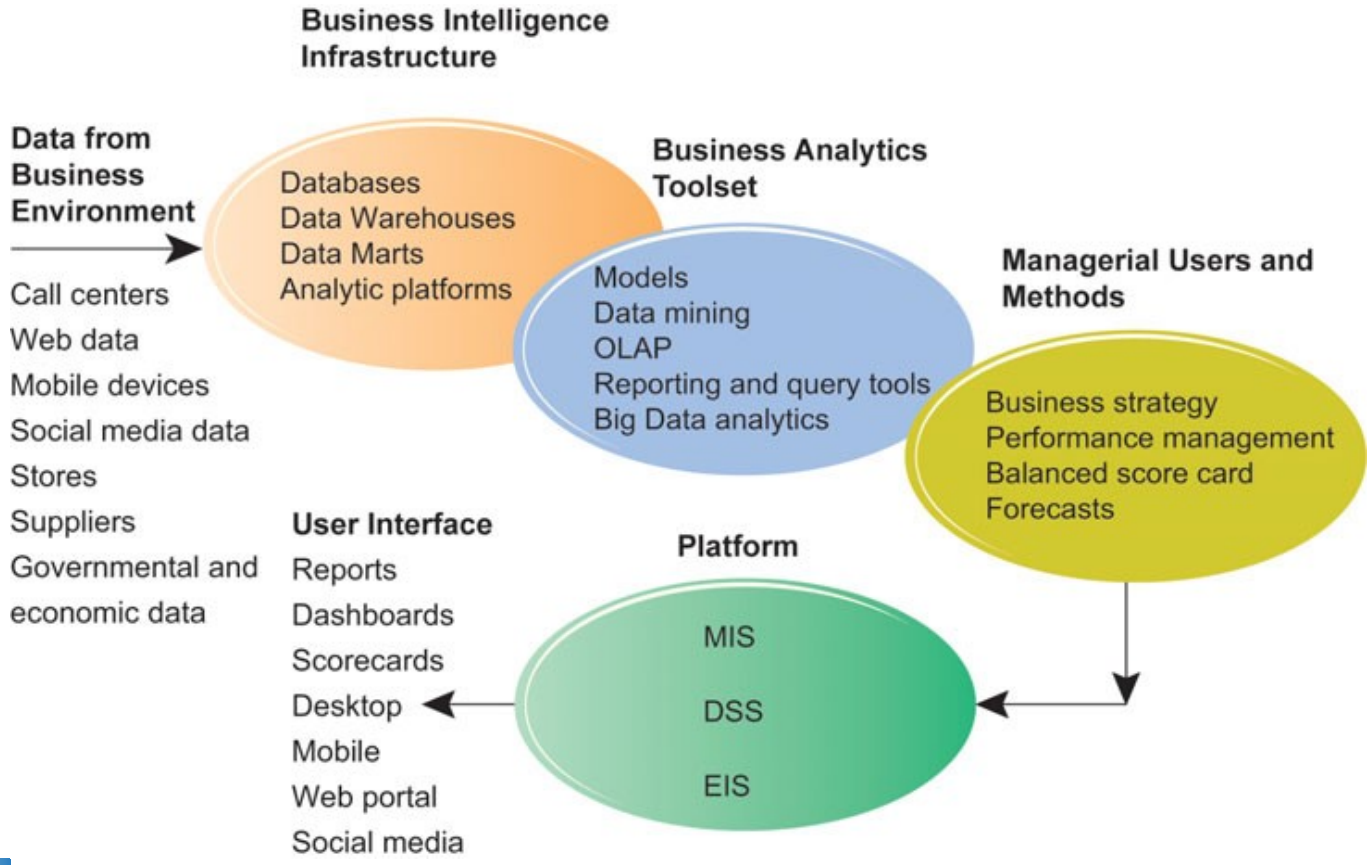
Arquitetura e Gestão de Sistemas de Informação

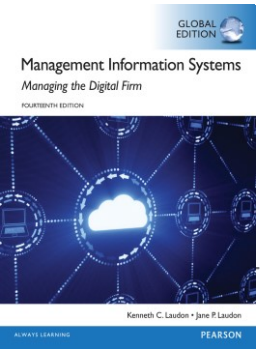
…examples of systems for both primary and support activities of a firm and of its value partners that can add a margin of value to a firm's products or services…





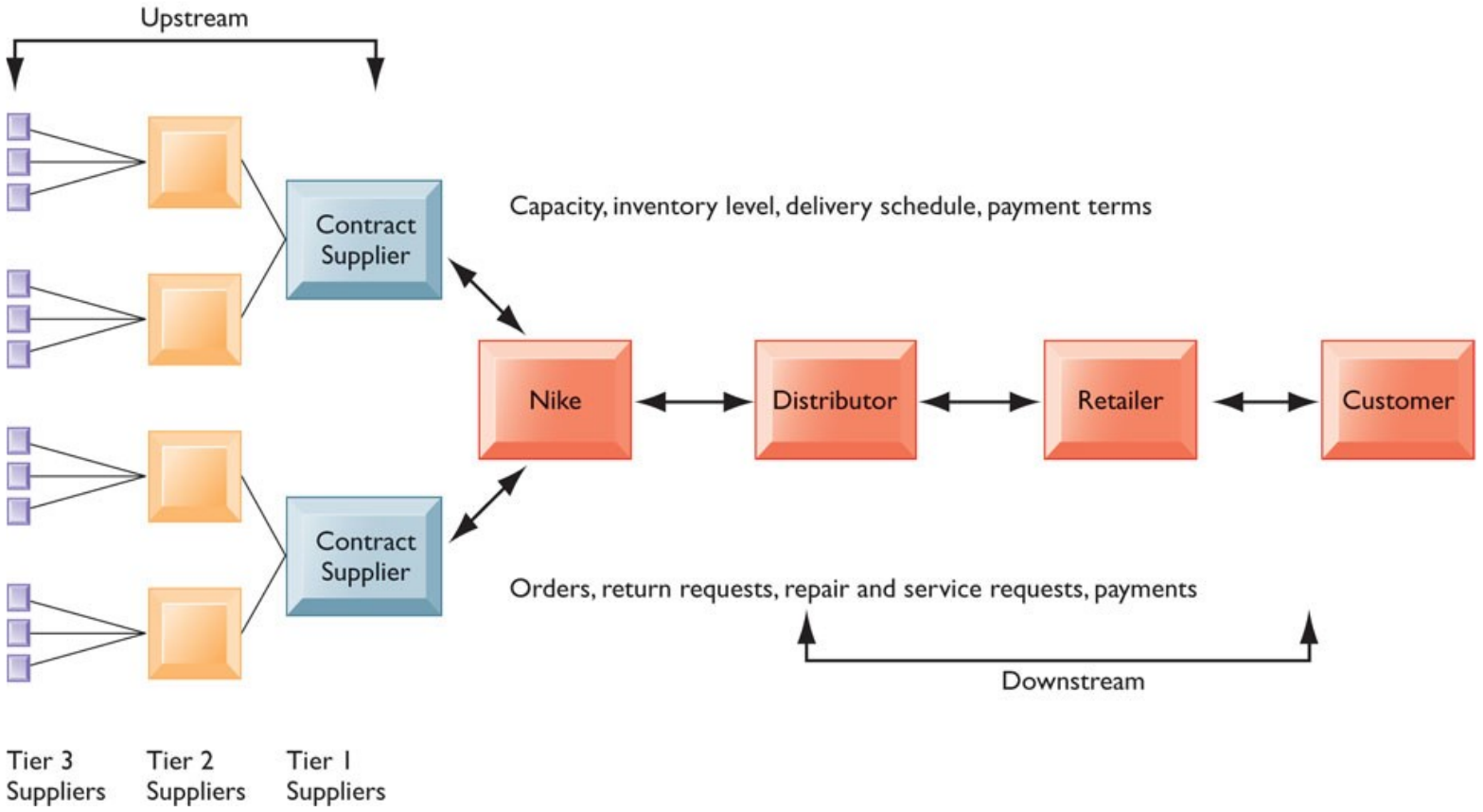
Arquitetura e Gestão de Sistemas de Informação

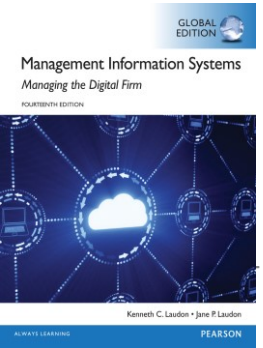




Business intelligence and **analytics** requires a strong database foundation, a set of analytic tools, and an involved management team that can ask intelligent questions and analyze data

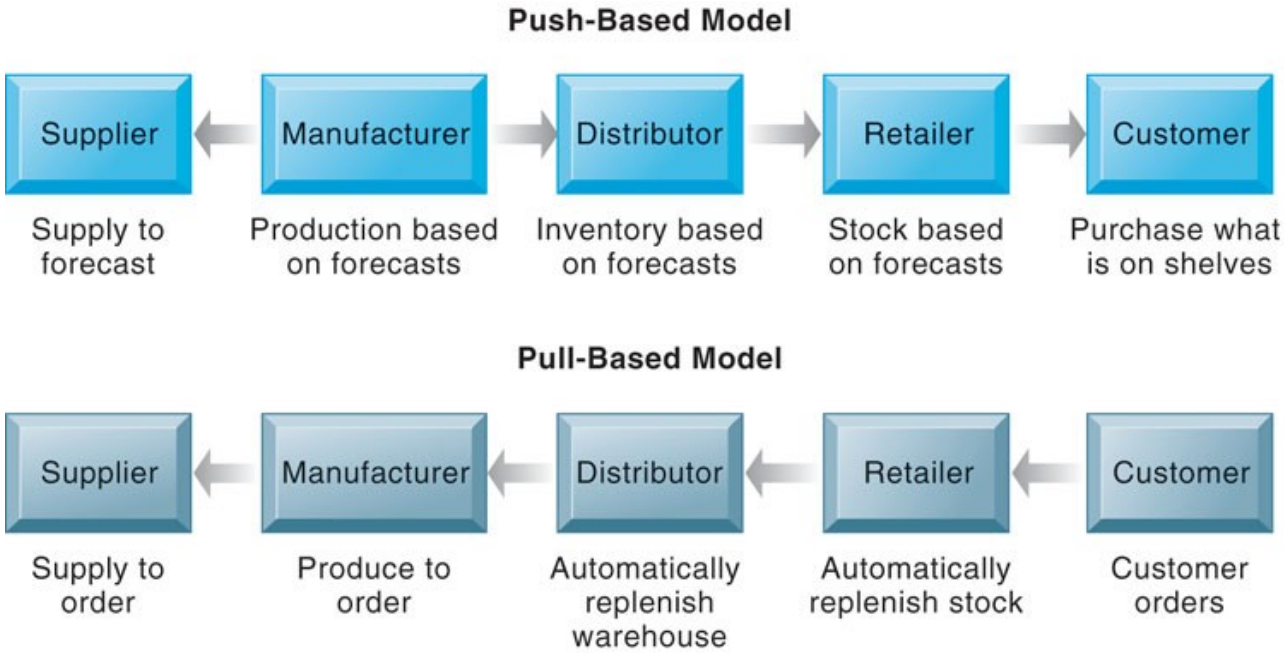
Arquitetura e Gestão de Sistemas de Informação

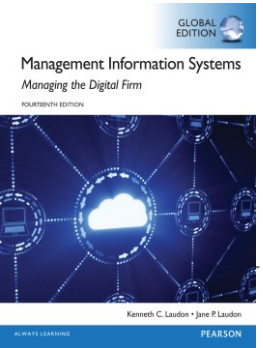




This figure illustrates the major entities in Nike's **supply chain** and the flow of information upstream and downstream to coordinate the activities involved in buying, making, and moving a product. Shown here is a simplified supply chain, with the upstream portion focusing only on the suppliers for sneakers and sneaker soles…

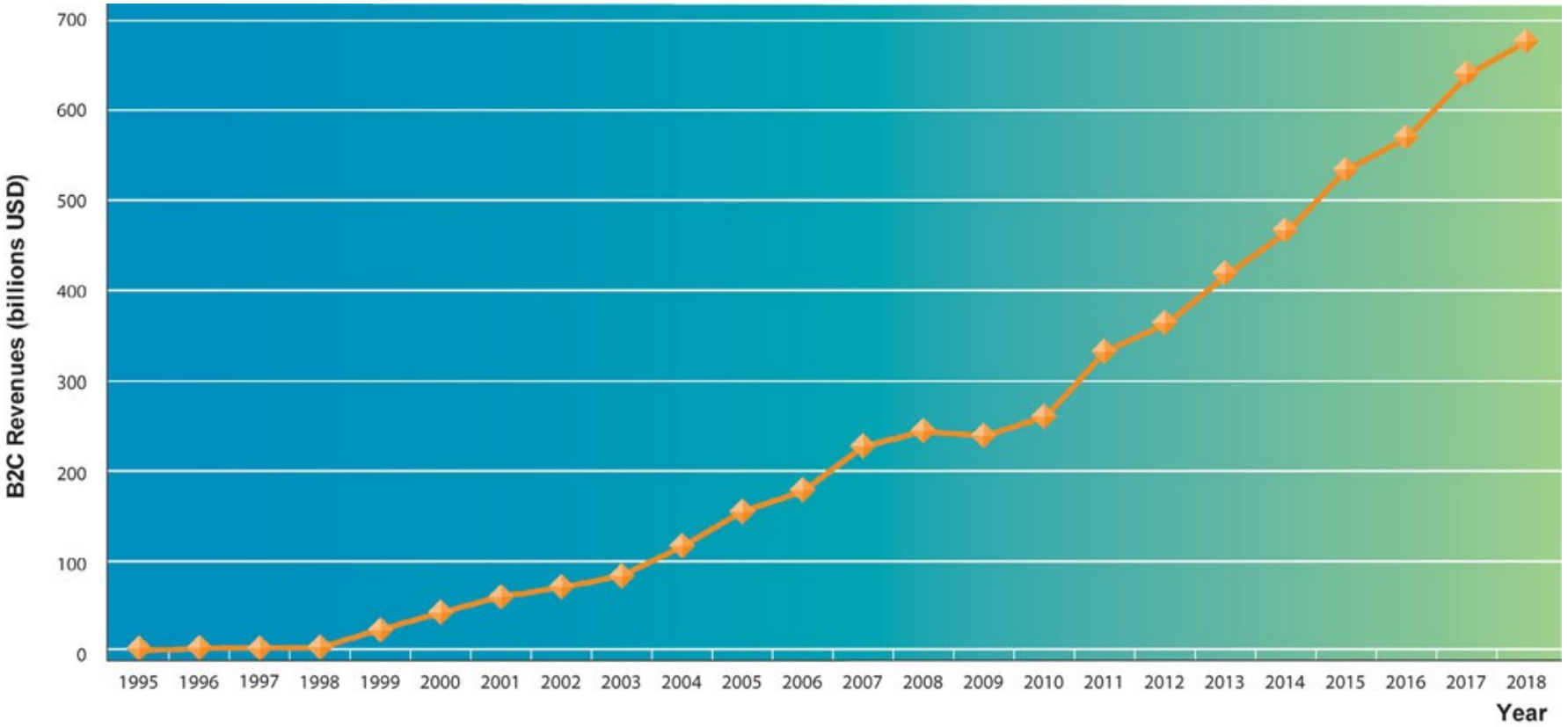
Arquitetura e Gestão de Sistemas de Informação

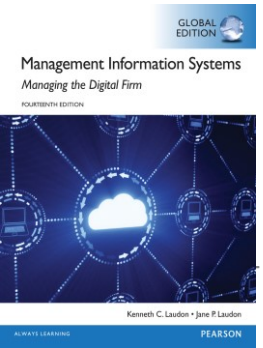




Supply Chains: The difference between push- and pull-based models is summarized by the slogan "Make what we sell, not sell what we make."

Arquitetura e Gestão de Sistemas de Informação

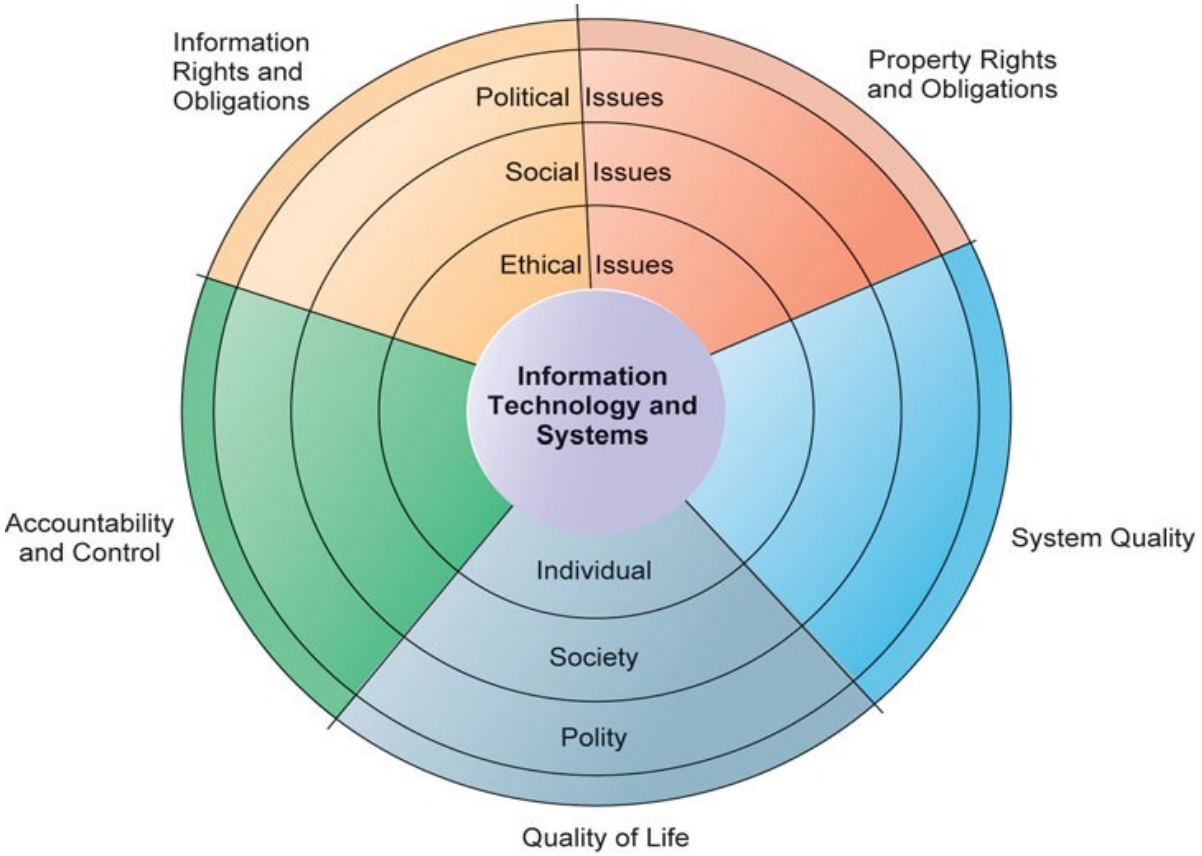


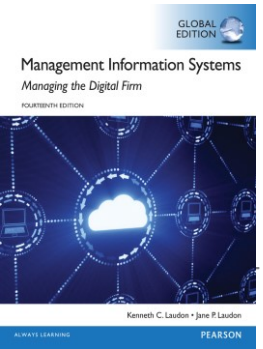


Retail **e-commerce** revenues grew 15–25 percent per year until the recession of 2008–2009, when they slowed measurably. In 2014, e-commerce revenues are growing again at an estimated 12 percent annually

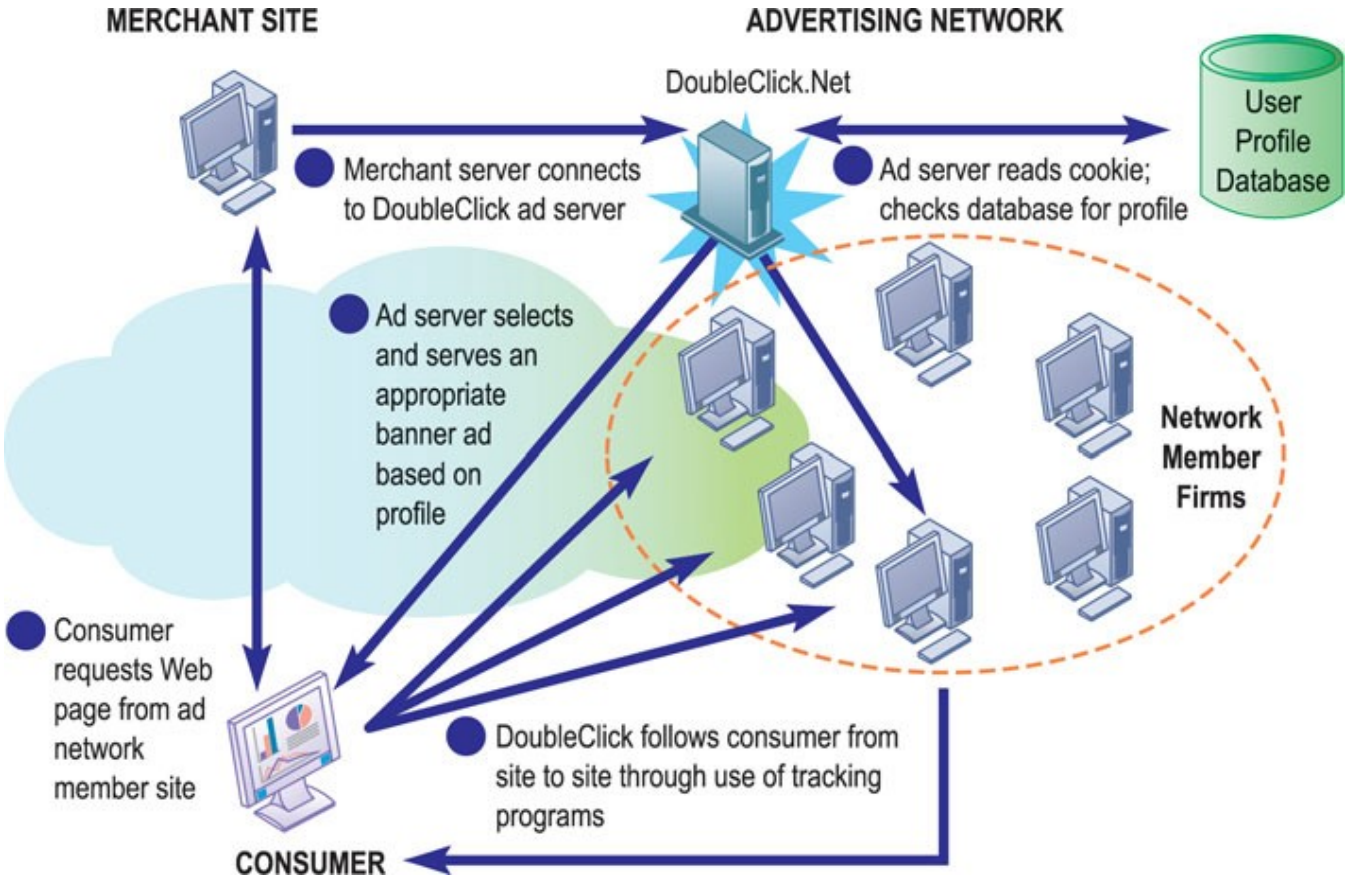
Arquitetura e Gestão de Sistemas de Informação

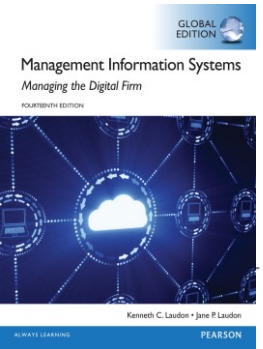
The introduction of new information technology has a ripple effect, raising new **ethical, social, and political issues** that must be dealt with on the individual, social, and political levels. These issues have **five moral dimensions: information rights and obligations, property rights and obligations, system quality, quality of life, and accountability and control**





Arquitetura e Gestão de Sistemas de Informação

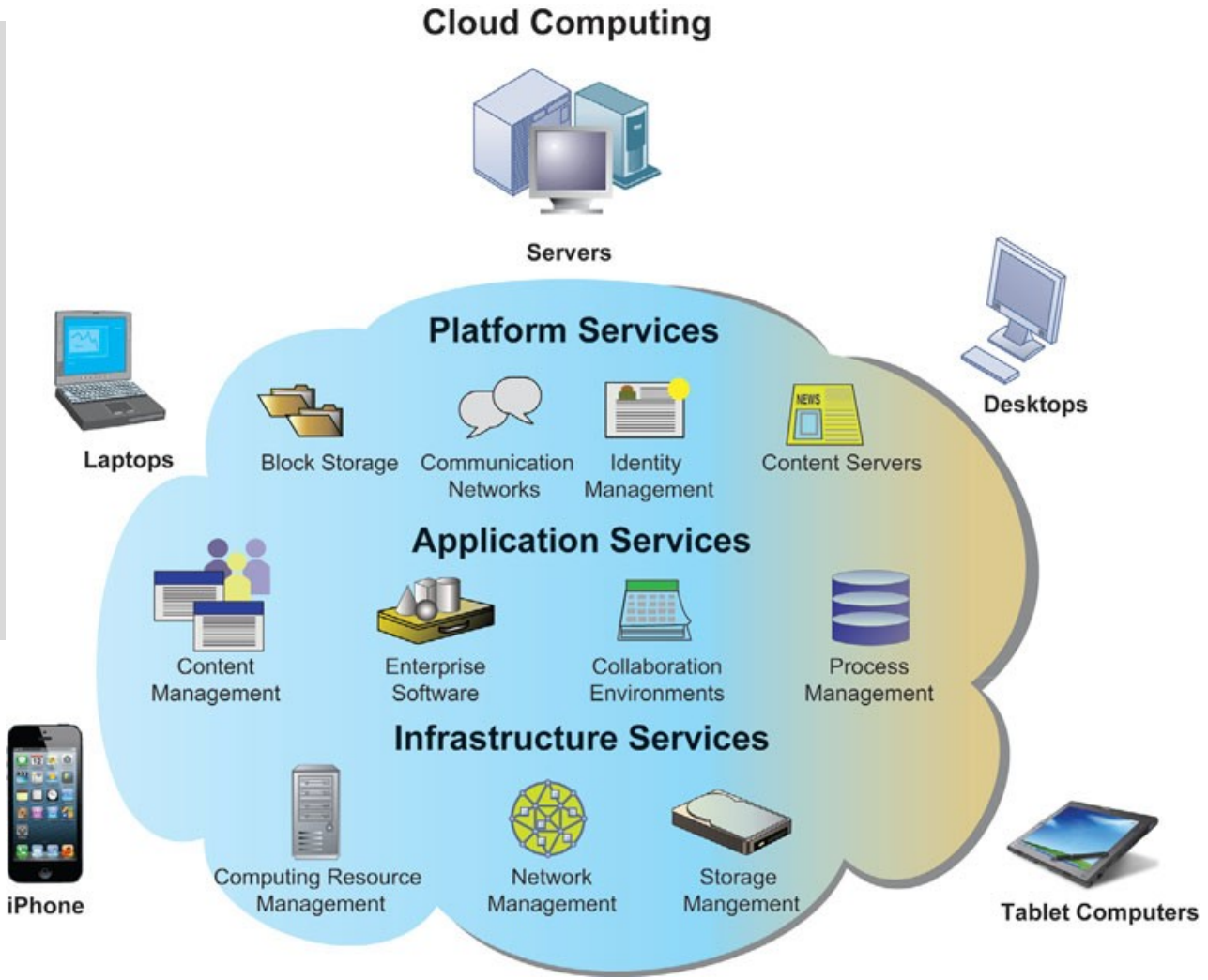


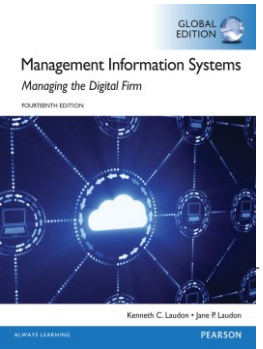


Advertising networks and their use of tracking programs have become **controversial** among privacy advocates because of their ability to track individual consumers across the Internet.

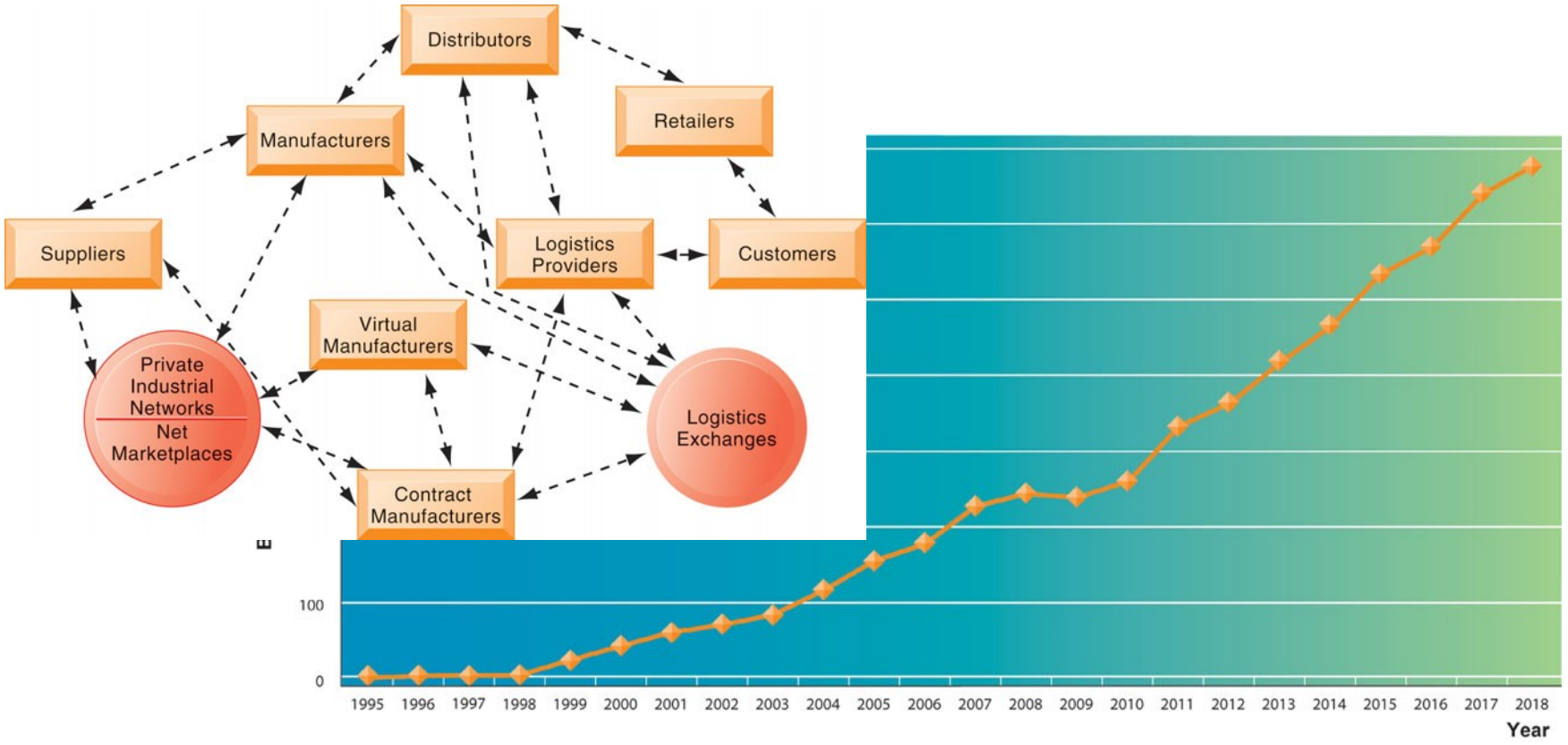
Arquitetura e Gestão de Sistemas de Informação

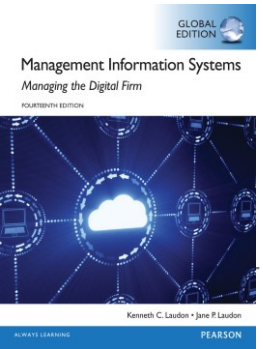
In cloud computing, hardware and software capabilities are a pool of virtualized resources provided over a network, often the Internet. Businesses and employees have access to applications and IT infrastructure anywhere, at any time, and on any device…





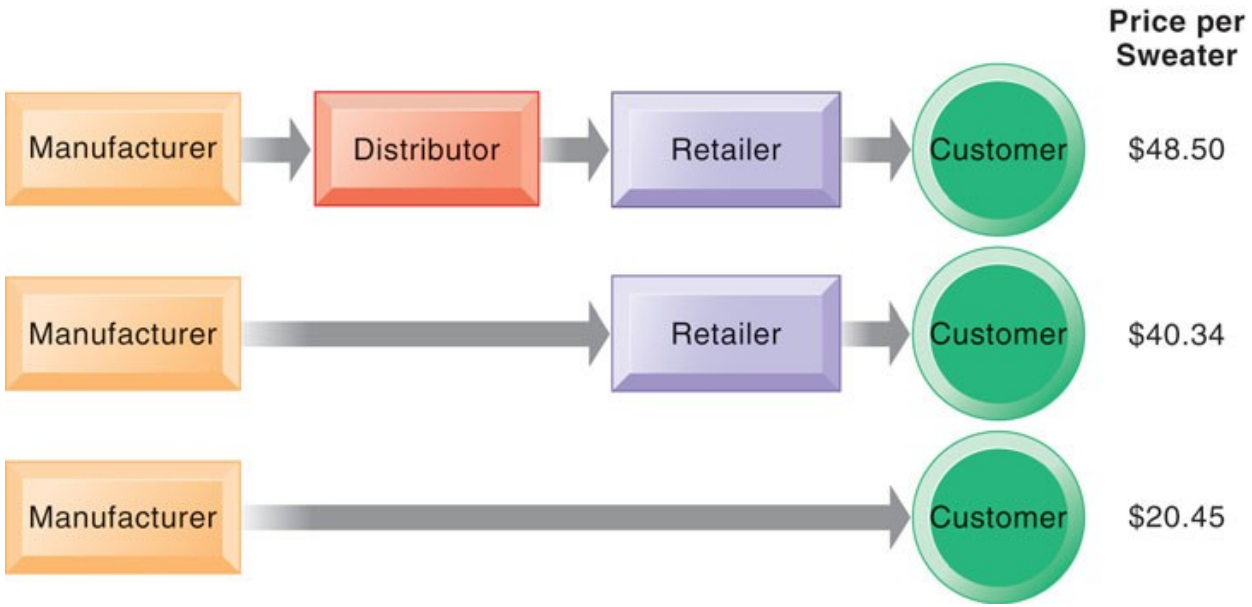
Arquitetura e Gestão de Sistemas de Informação



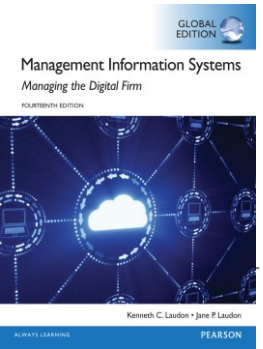


The emerging Internet-driven **supply chain** operates like a digital logistics nervous system. It provides multidirectional communication among firms, networks of firms, and e-marketplaces so that entire networks of supply chain partners can immediately adjust inventories, orders, and capacities.

Arquitetura e Gestão de Sistemas de Informação



The typical distribution channel has several intermediary layers, each of which adds to the final cost of a product, such as a sweater. Removing layers lowers the final cost to the customer.



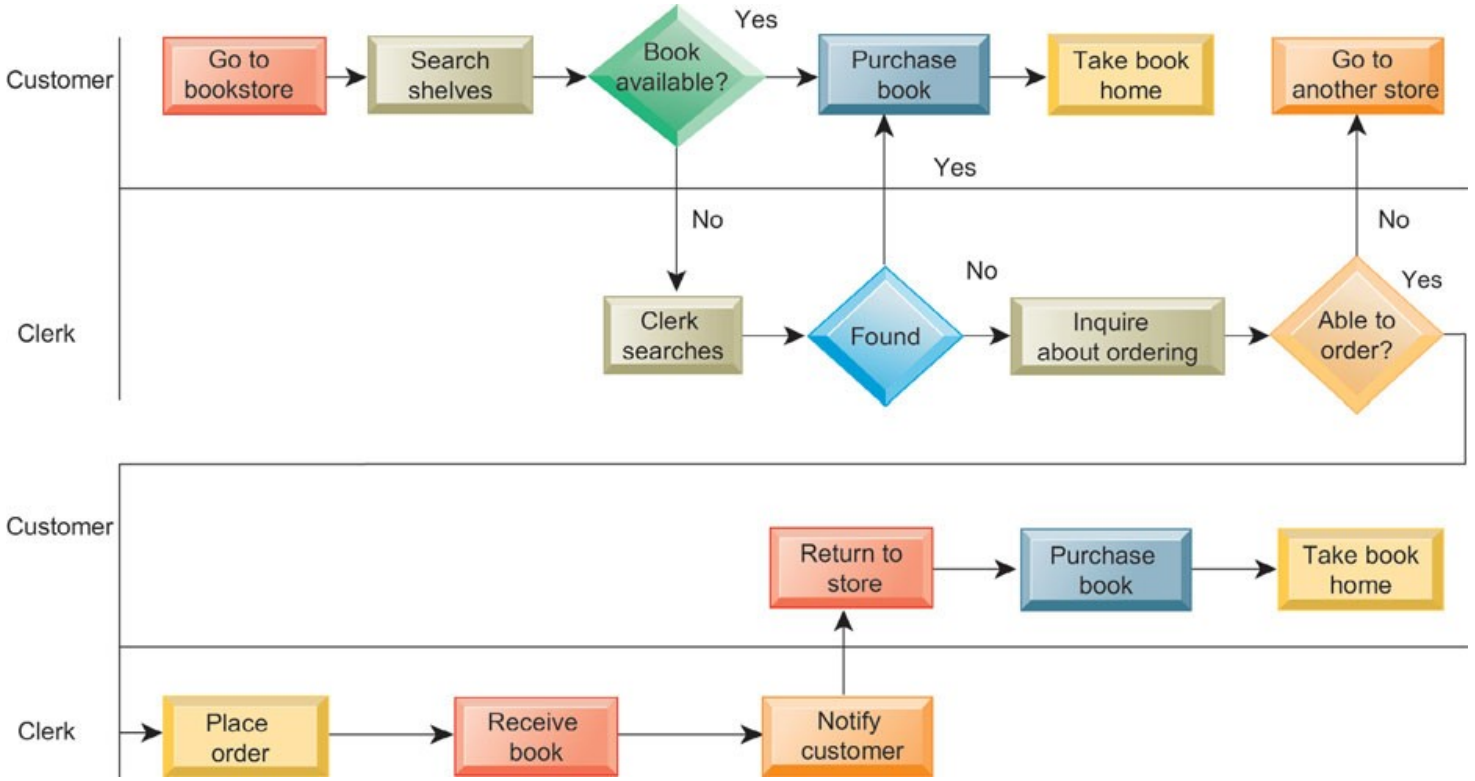
Business Process

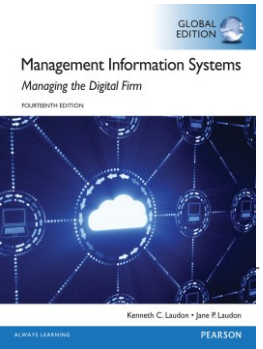
- Set of interrelated activities that transform inputs into outputs in order to produce a service or product to a specific customer.
- The focus is always on the final product.

- A business process is usually abstracted as a transformational process
  - IPO (Input – Process – Output)
  - There are other paradigms

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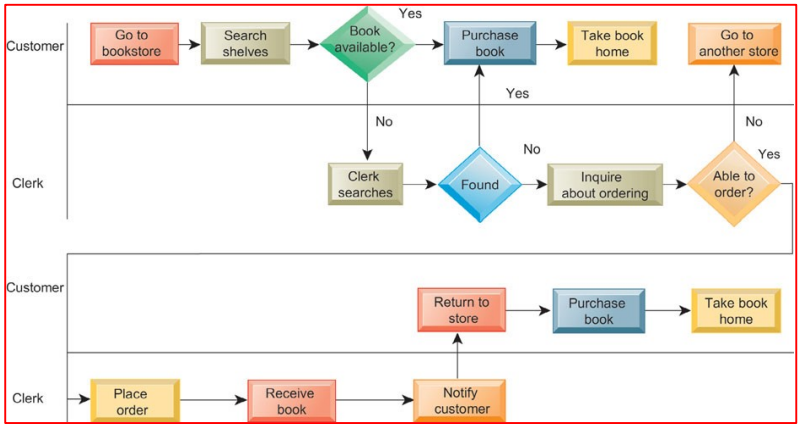
Arquitetura e Gestão de Sistemas de Informação

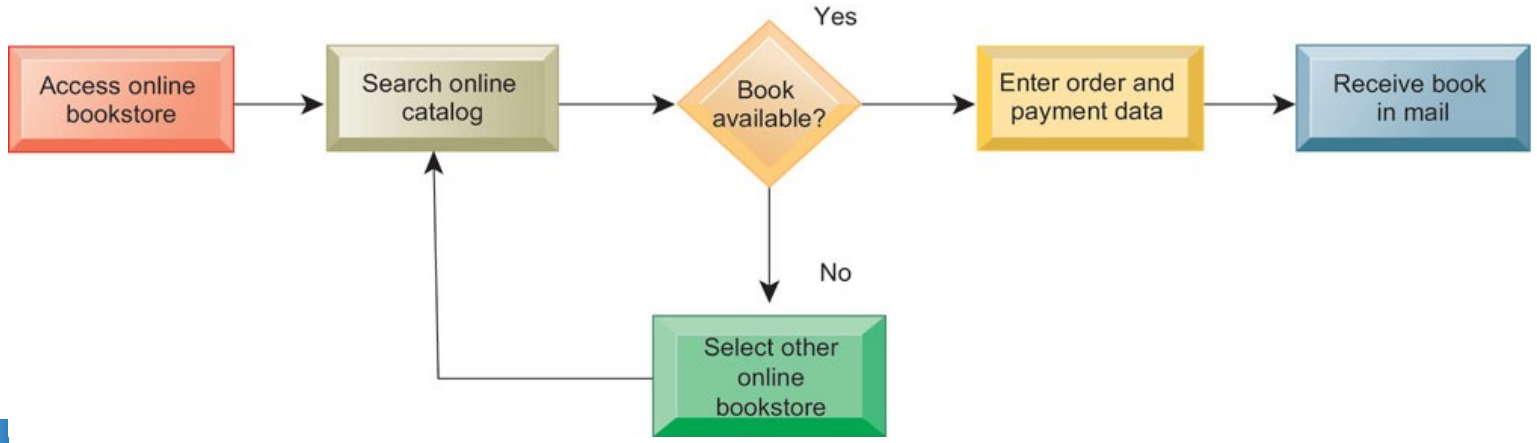


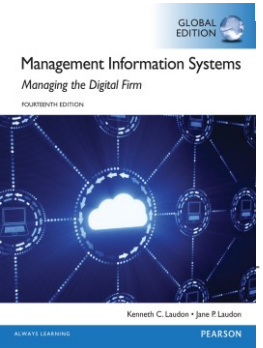


Purchasing a book from a physical bookstore requires many steps to be performed by both the seller and the customer.

Arquitetura e Gestão de Sistemas de Informação



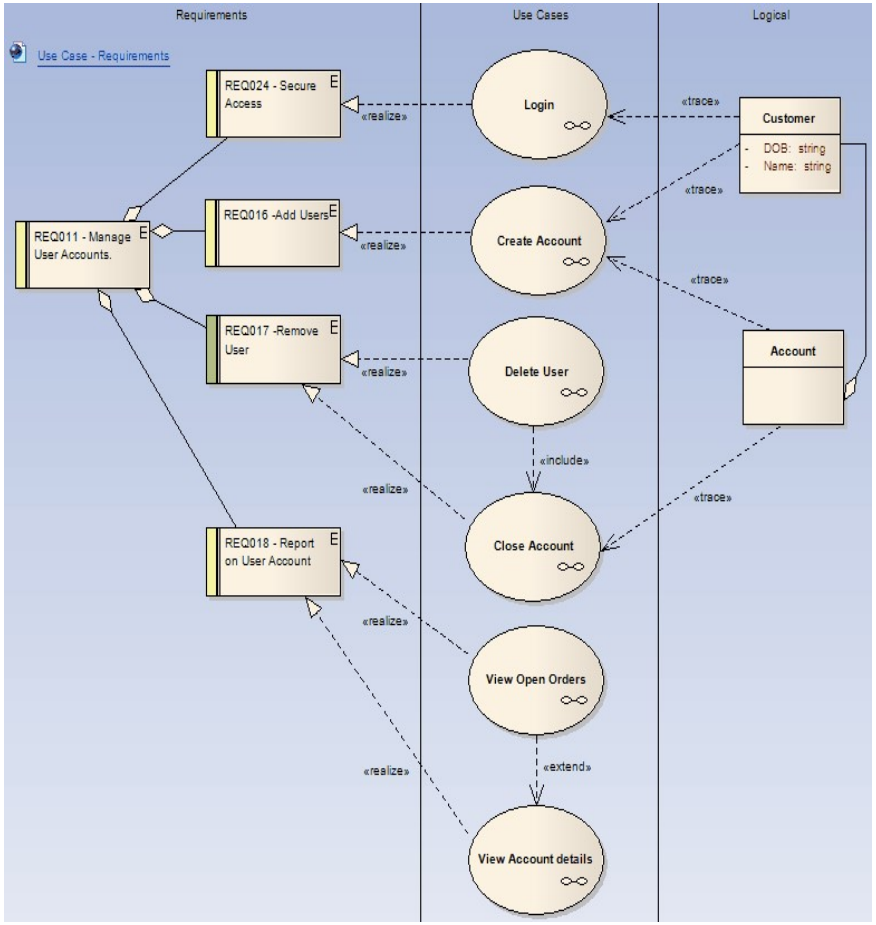


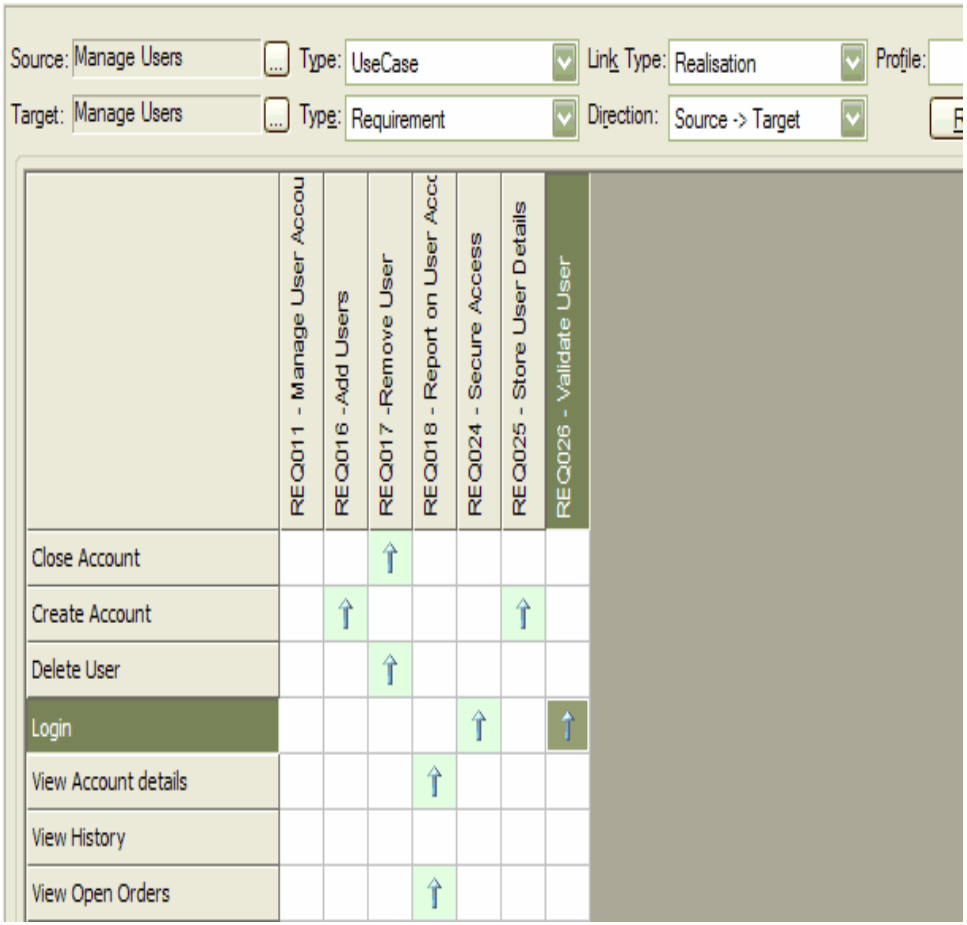


Using Internet technology makes it possible to redesign the process for purchasing a book so that it requires fewer steps and consumes fewer resources.

Desenho de Sistemas – Ratreabilidade (Traceability)

Traceability is a fundamental technique to relate decisions with their origin and/or implications.





What is a Requirement?

Definition 2-1: *Requirement*

(1) A condition or capability needed by a user to solve a problem or achieve an objective.
(2) A condition or capability that must be met or possessed by a system or system component to satisfy a contract, standard, specification, or other formally imposed documents.
(3) A documented representation of a condition or capability as in (1) or (2).

[IEEE Std 610.12-1990]

Definition 2-2: *Requirements artefact*

A requirements artefact is a documented requirement.

Main Types of Requirements

- Functional Requirements

Definition 2-3: *Functional requirements*

"These [functional requirements] are statements of services the system should provide, how the system should react to particular inputs and how the system should behave in particular situations. In some cases, the functional requirements may also state what the system should not do. [...]

[...] When expressed as user requirements, the requirements are usually described in a fairly abstract way. However, functional system requirements describe the system function in detail, its inputs and outputs, exceptions, and so on."

[Sommerville 2007]

Example 2-2: Functional requirement

R31 If a sensor detects that a glass pane is damaged or broken, the system shall inform the security company.

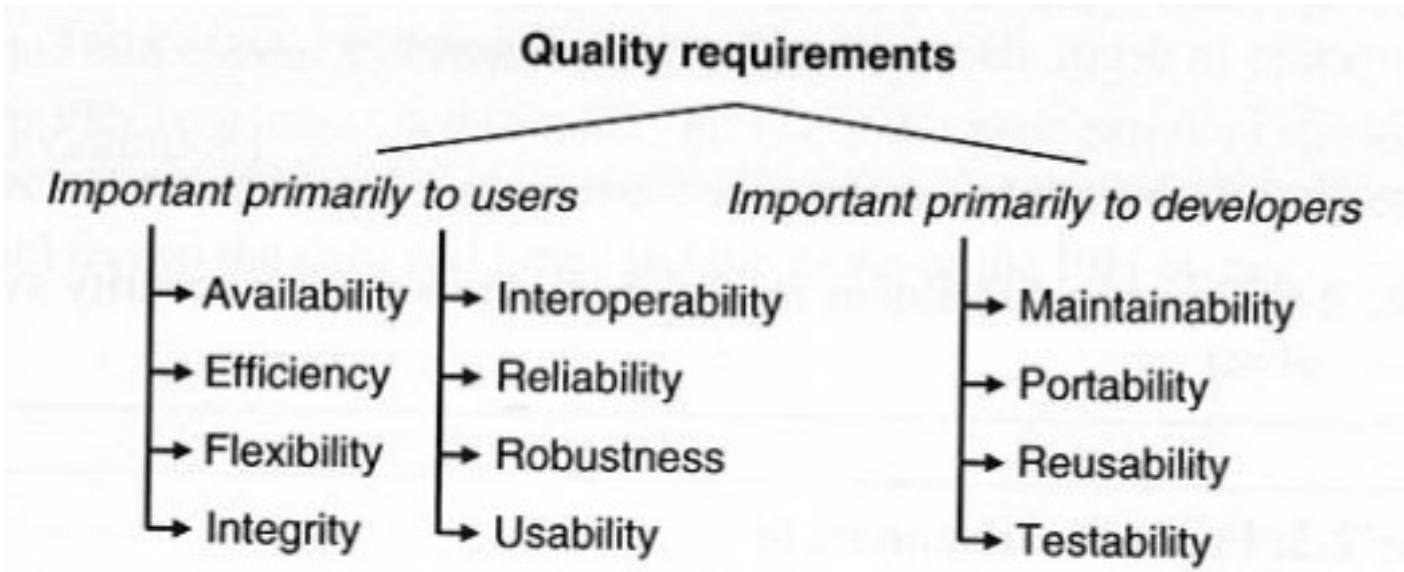
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Main Types of Requirements

- Quality Requirements

Definition 2-4: *Quality requirement*

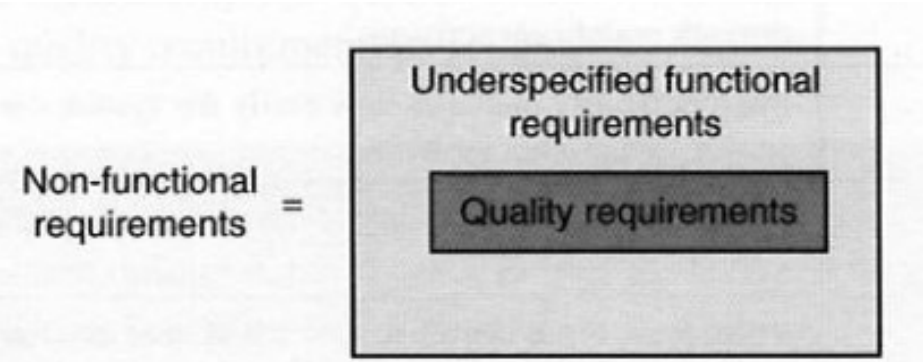
A quality requirement defines a quality property of the entire system or of a system component, service, or function.



A quality requirements taxonomy, as defined in [Wiegers 2003]

Main Types of Requirements

- Non-Functional Requirements



Example 2-3: Non-functional requirement/underspecified functional requirement

R12 The system shall be secure.

On looking at requirement R12 from Example 2-3 in slightly more detail, several questions arise due to the underspecification of this requirement, such as:

- What does the adjective "secure" mean?
- Which properties shall the system provide in order to be "secure"?
- How can one check whether the implemented system is "secure"?

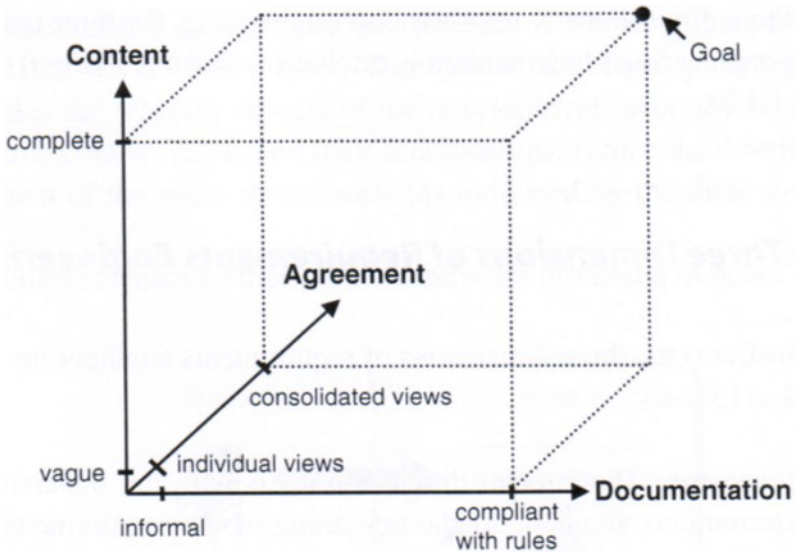
Goals of Requirements Engineering



Definition 4-1: *Goals of requirements engineering*

Requirements engineering is a cooperative, iterative, and incremental process which aims at ensuring that:

(1) All relevant requirements are explicitly known and understood at the required level of detail.
(2) A sufficient agreement about the system requirements is achieved between the stakeholders involved.
(3) All requirements are documented and specified in compliance with the relevant documentation/specification formats and rules.



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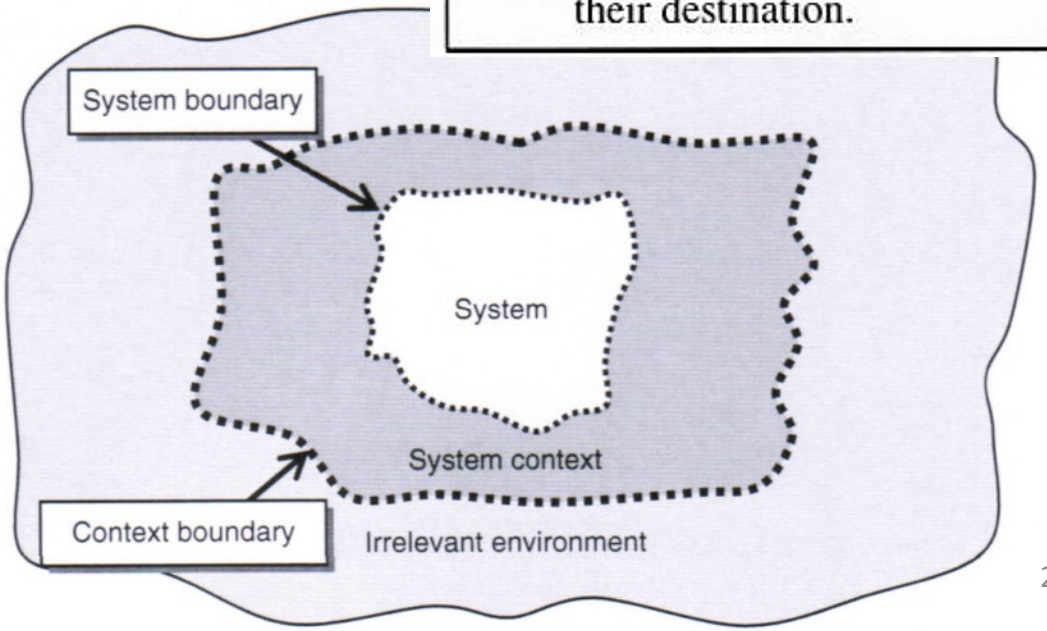
Requirements and System context



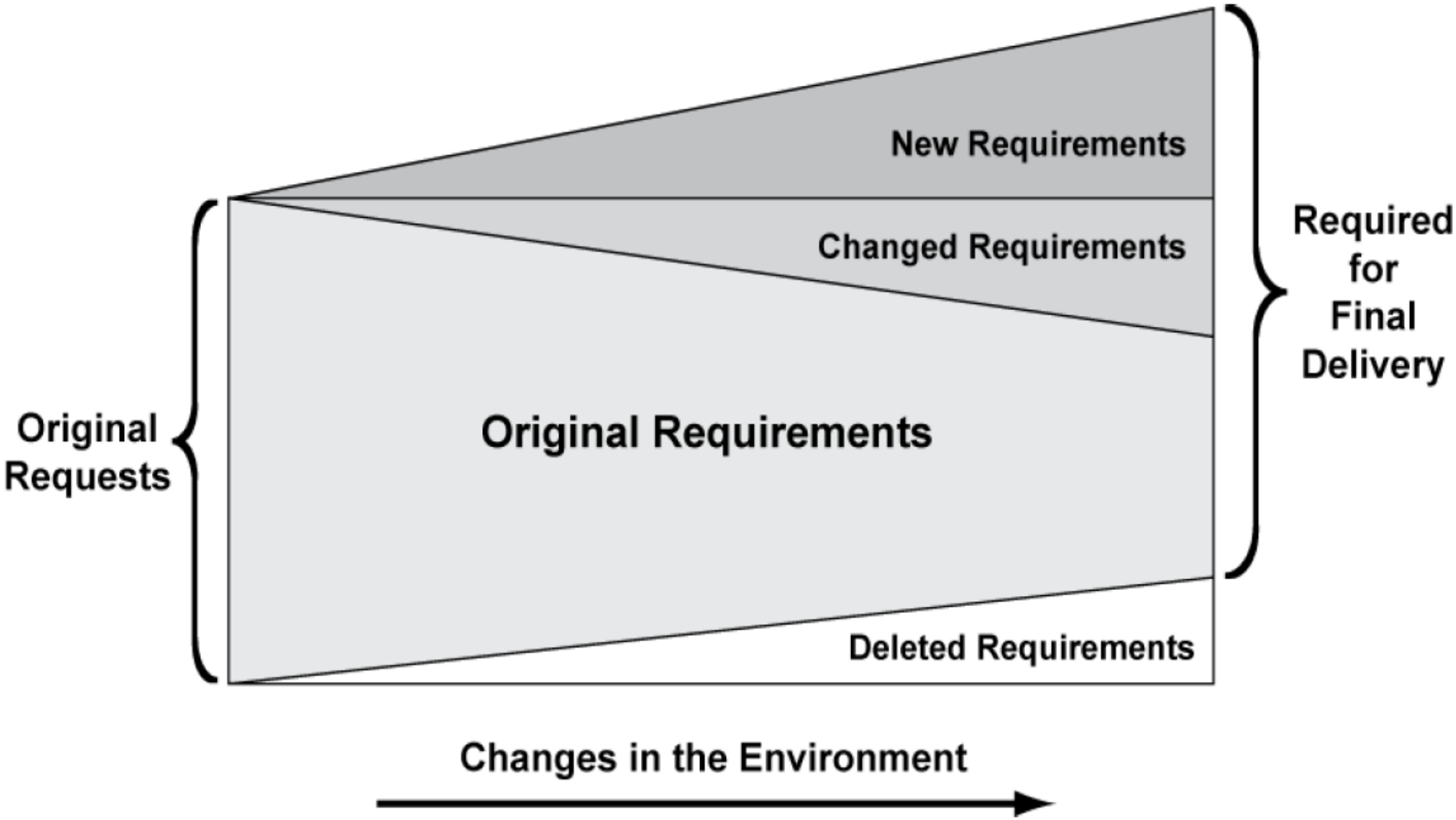
Requirements are defined and interpreted within a given context. A change in the context can change the meaning of a requirement dramatically. For example, the way a requirement is interpreted is significantly influenced by the documented context information. If insufficient or even no context information is documented, the same requirement can be interpreted in fairly different ways.

Example 5-4: Documented requirement with vague (almost no) context information

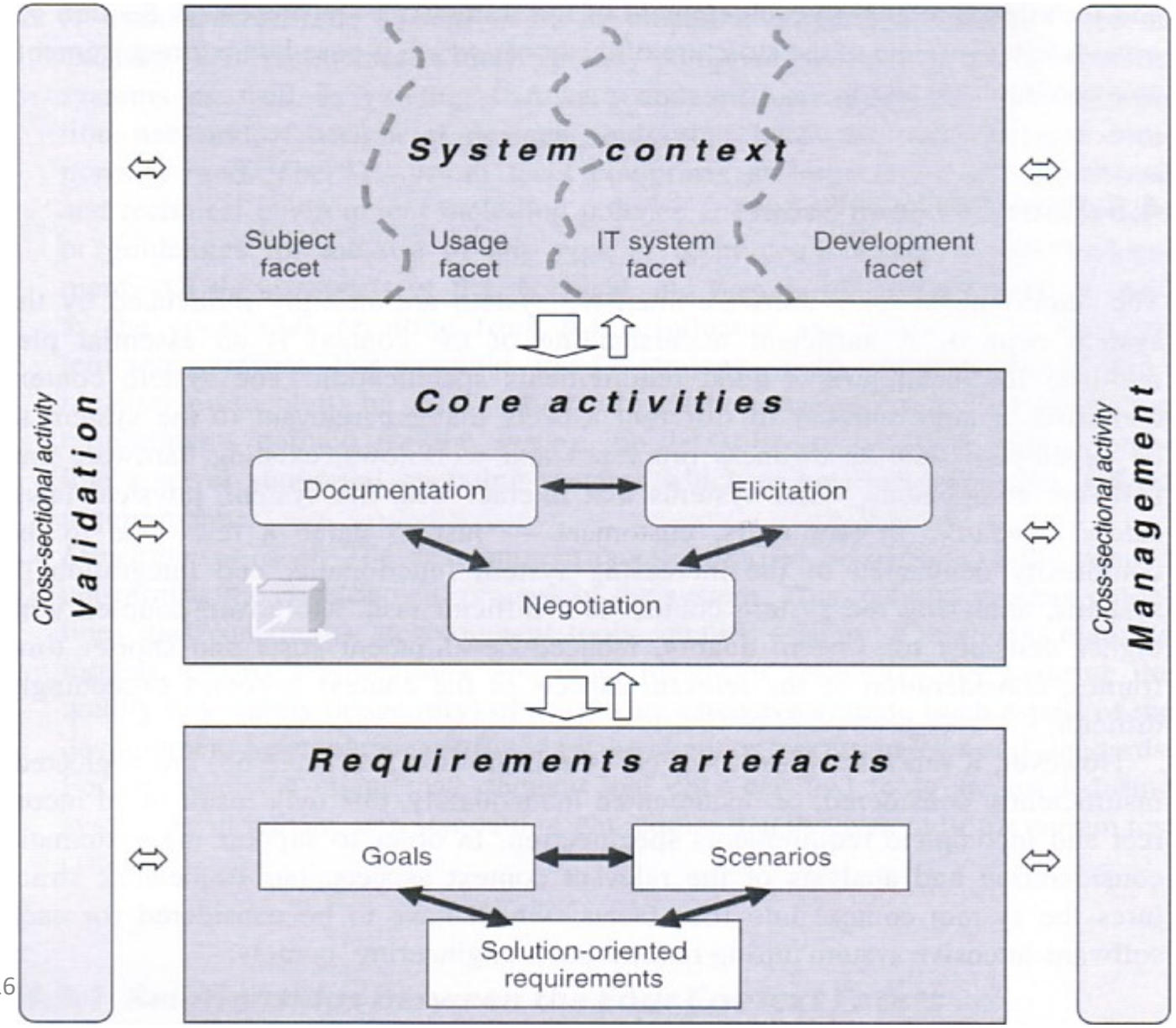
R23: The planned means of transportation shall offer travellers a fast journey to their destination.



Requirements change over time ☹



A Requirements Engineering Framework



Facet ≈ Viewpoint

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Requirements Models



Definition 20-1: *Requirements model*

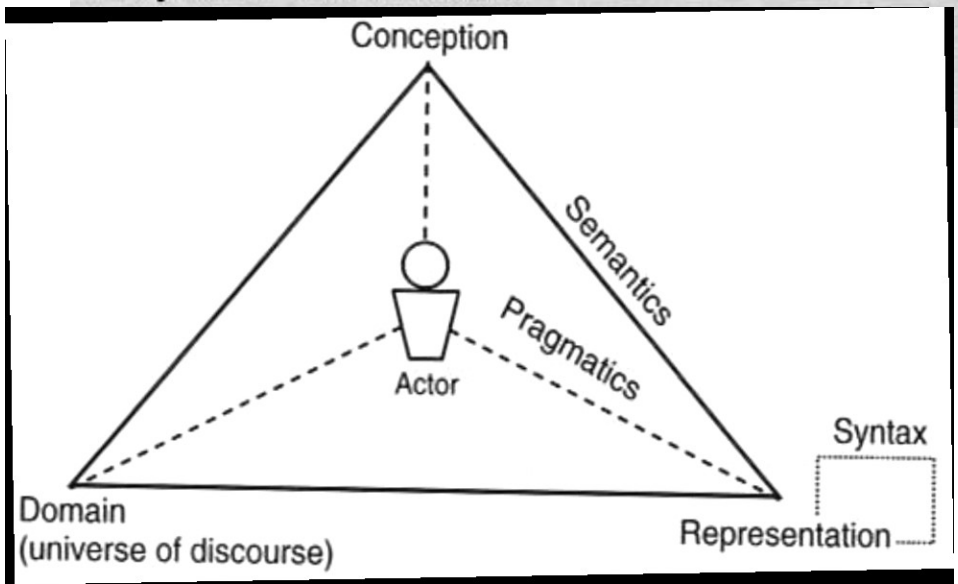
Requirements models are conceptual models that document requirements (goals, scenarios, or solution-oriented requirements) using a conceptual modelling language.

Definition 19-2: *Model*

A model is an abstract representation of the universe of discourse created for a specific purpose (use).

Definition 19-1: *Universe of discourse*

The universe of discourse comprises any part or aspect of the existing or conceived reality under consideration.



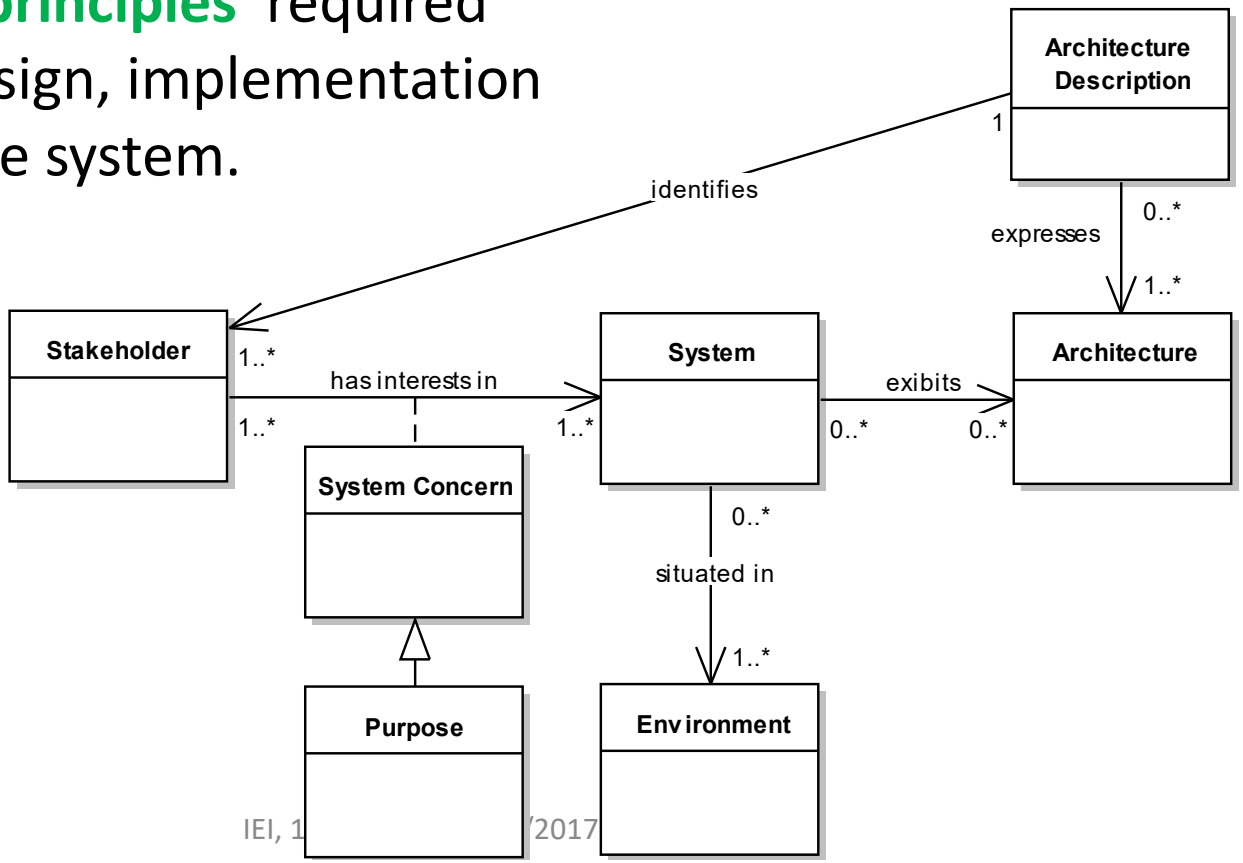
based on [Frisco 1998]

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System Architecture

- A [system] **architecture** is a formal representation of a system that defines the **structure** and the properties of its components, as well as the relationships between them (i.e the **behaviour**) .

- It also defines the **principles** required for the analysis, design, implementation and evolution of the system.



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Structure & Behaviour

Structure and behaviour are two general ways of classifying **viewpoints**.

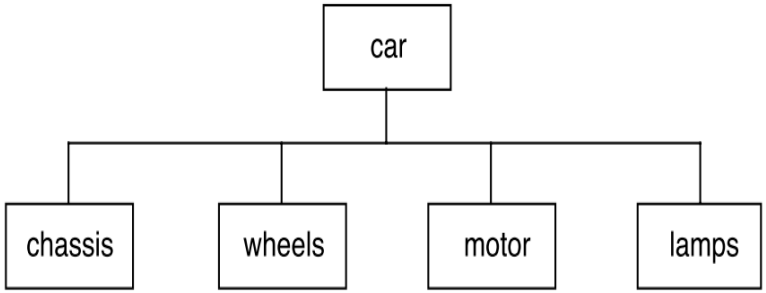
So, why are they so relevant?

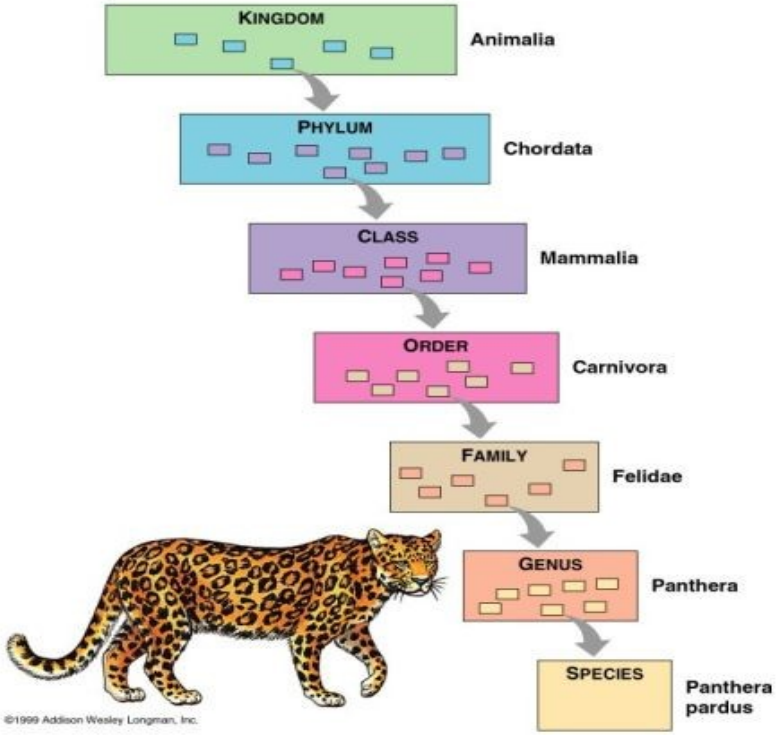
We assume that structure and behaviour are related to concerns that most of the time can pragmatically be separated orthogonally.

[cf. orthogonal separation of concerns]

Structure

Structure is a fundamental notion referring to the recognition, observation, nature, and stability of patterns and relationships of entities.



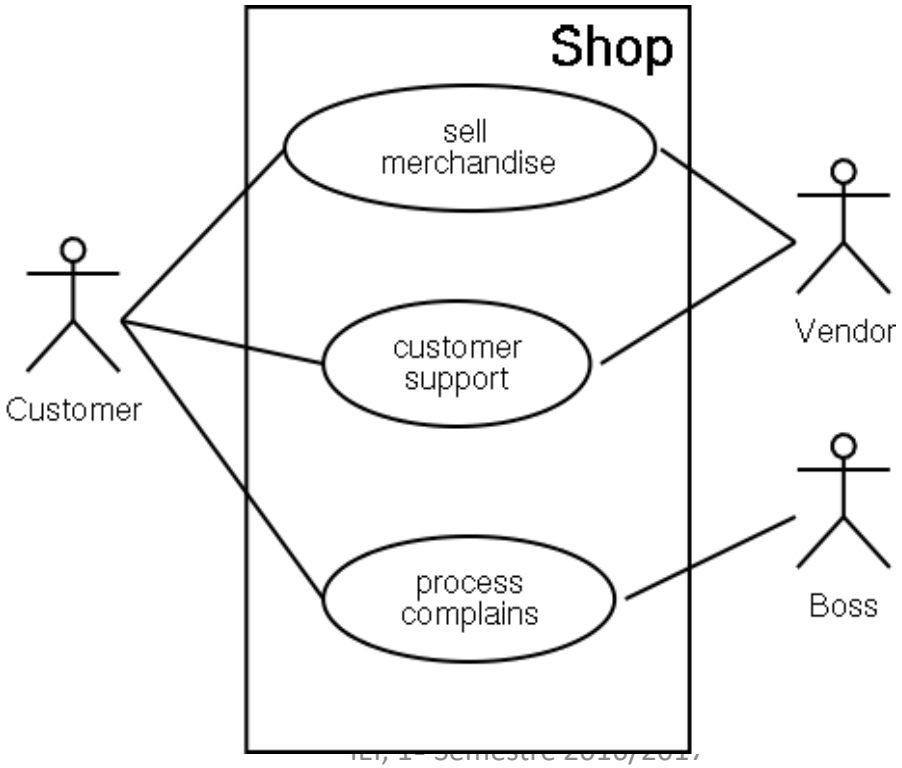


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Behaviour

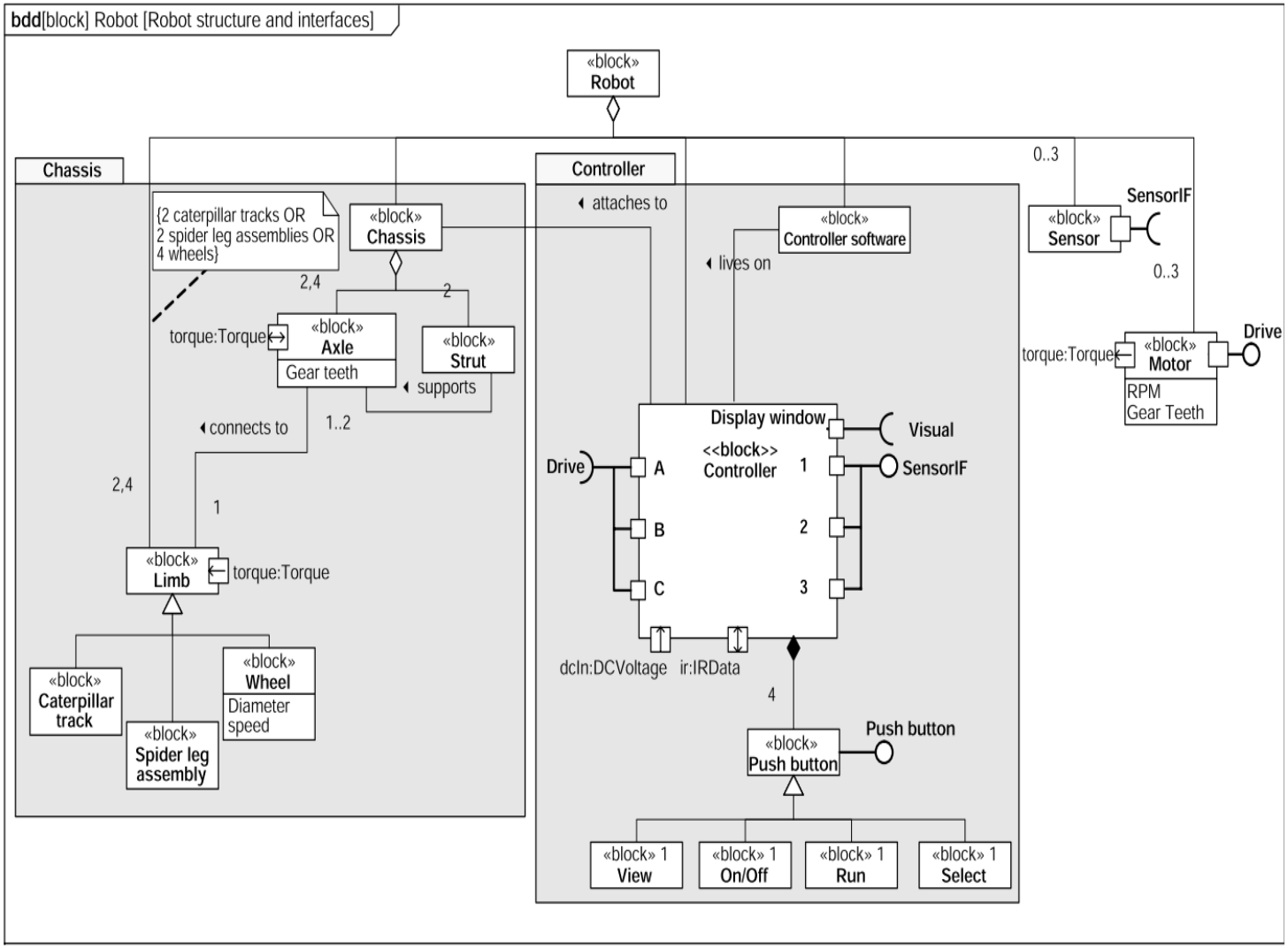
Behaviour refers to the actions of a system.

It is the response of the system to various stimuli (or inputs), whether internal or external.



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Structure (SysML diagram)



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Behaviour (BPMN diagram)

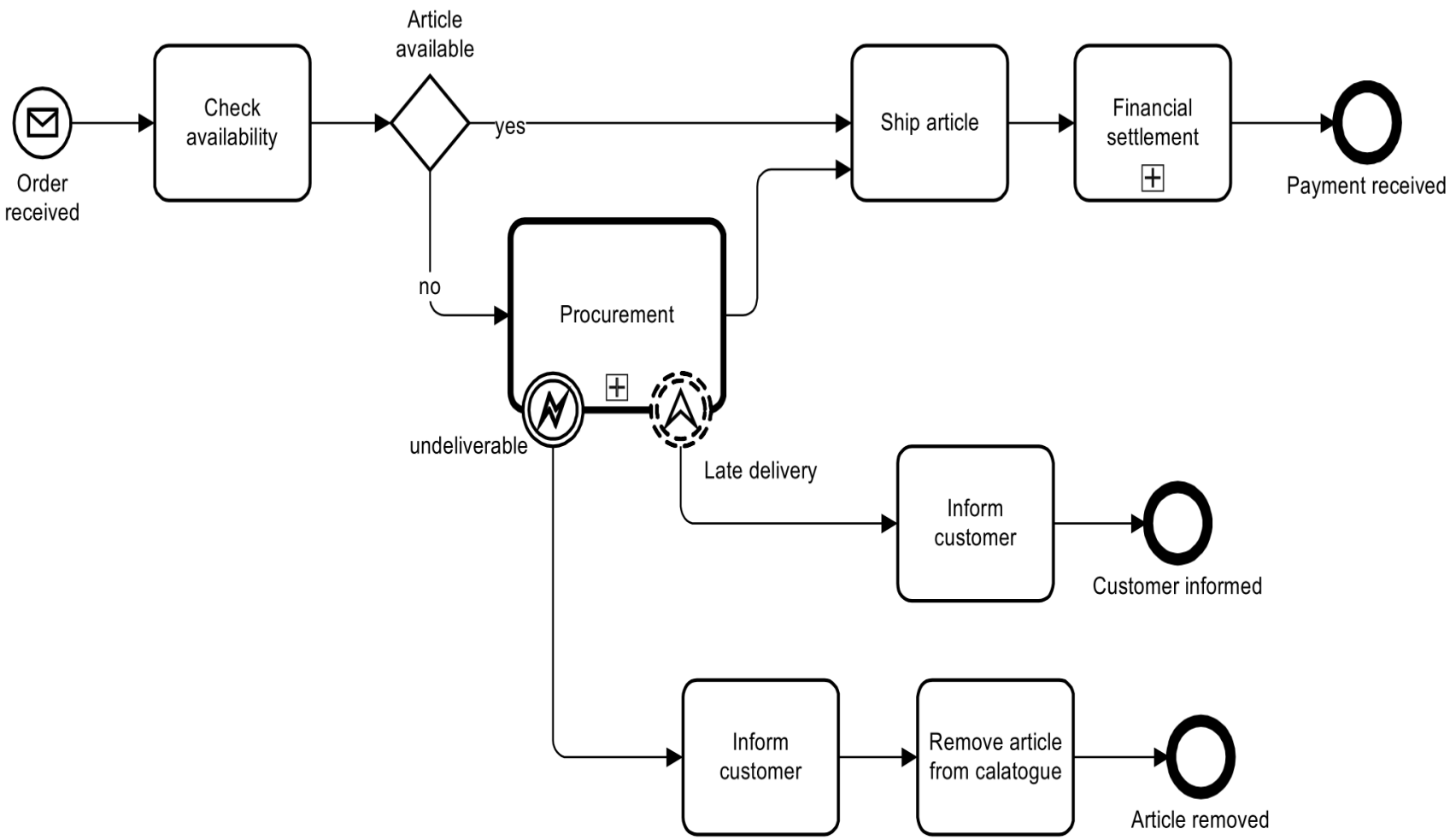


Figure 5.3: Order Fulfillment

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Modelação de Sistemas de Informação

- ArchiMate is a technical standard from The Open Group and is based on the concepts of the IEEE 1471 standard (now ISO 42010). It is supported by various tool vendors and consulting firms.
- ArchiMate distinguishes itself from other languages such as Unified Modeling Language (UML) and Business Process Modeling and Notation (BPMN) by its enterprise modelling scope.
- The **Unified Modeling Language** (**UML**) is a general-purpose, developmental, modeling language in the field of software engineering, that is intended to provide a standard way to visualize the design of a system.
- In 1997 UML was adopted as a standard by the Object Management Group (OMG), and has been managed by this organization ever since. In 2005 UML was also published by the International Organization for Standardization (ISO) as an approved ISO standard.
- **Business Process Model and Notation** (**BPMN**) is a graphical representation for specifying business processes in a business process model.
- Business Process Management Initiative (BPMI) developed BPMN, which has been maintained by the Object Management Group since the two organizations merged in 2005. Version 2.0 of BPMN was released in January 2011, at which point the name was adapted to **Business Process Model *and* Notation** as execution semantics were also introduced alongside the notational and diagramming elements.

- https://en.wikipedia.org/wiki/ArchiMate
- https://en.wikipedia.org/wiki/Unified_Modeling_Language
- https://en.wikipedia.org/wiki/Business_Process_Model_and_Notation

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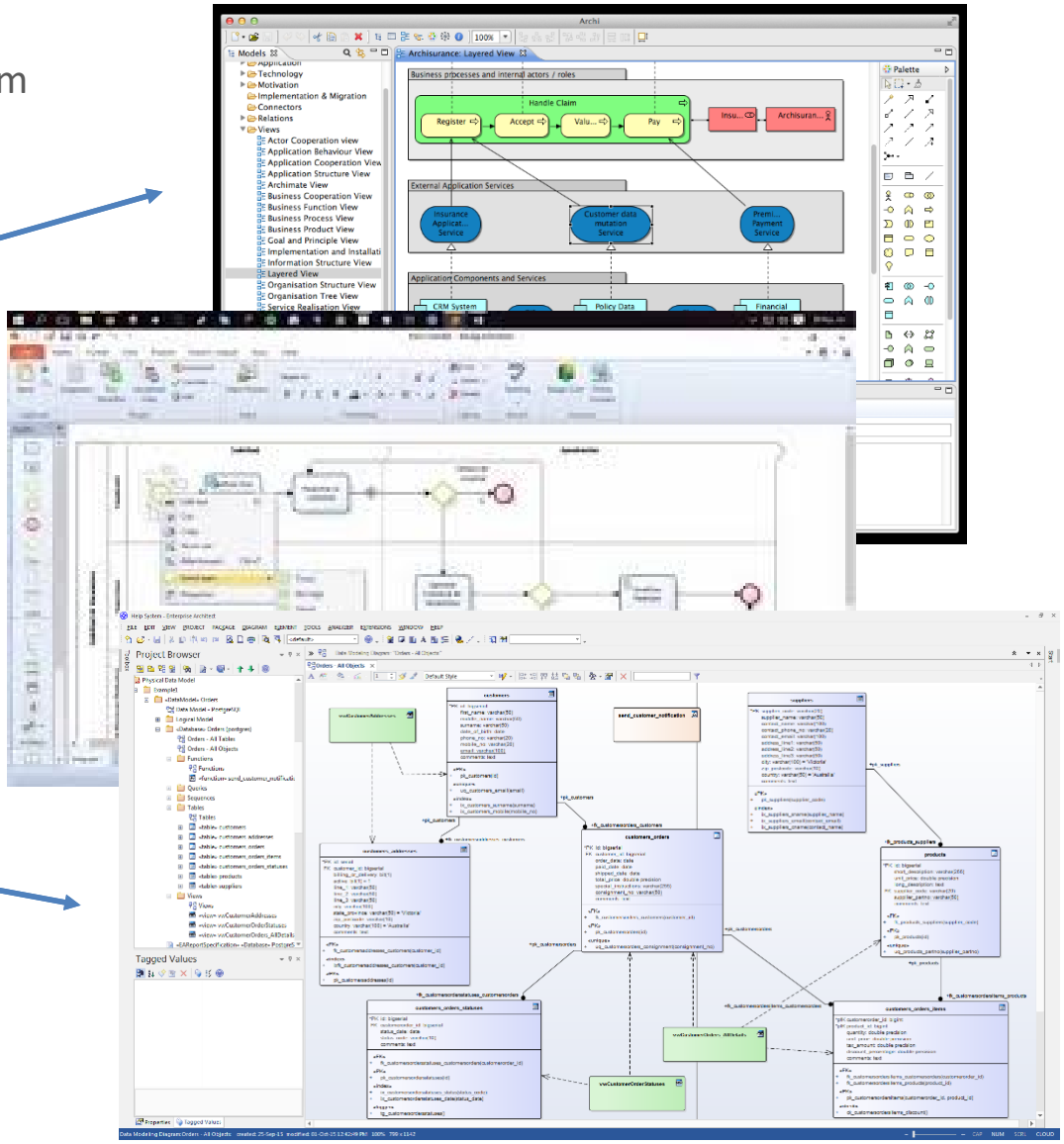
Modelação de Sistemas de Informação

Examples of diagrams from modeling languages:

ArchiMate

BPMN

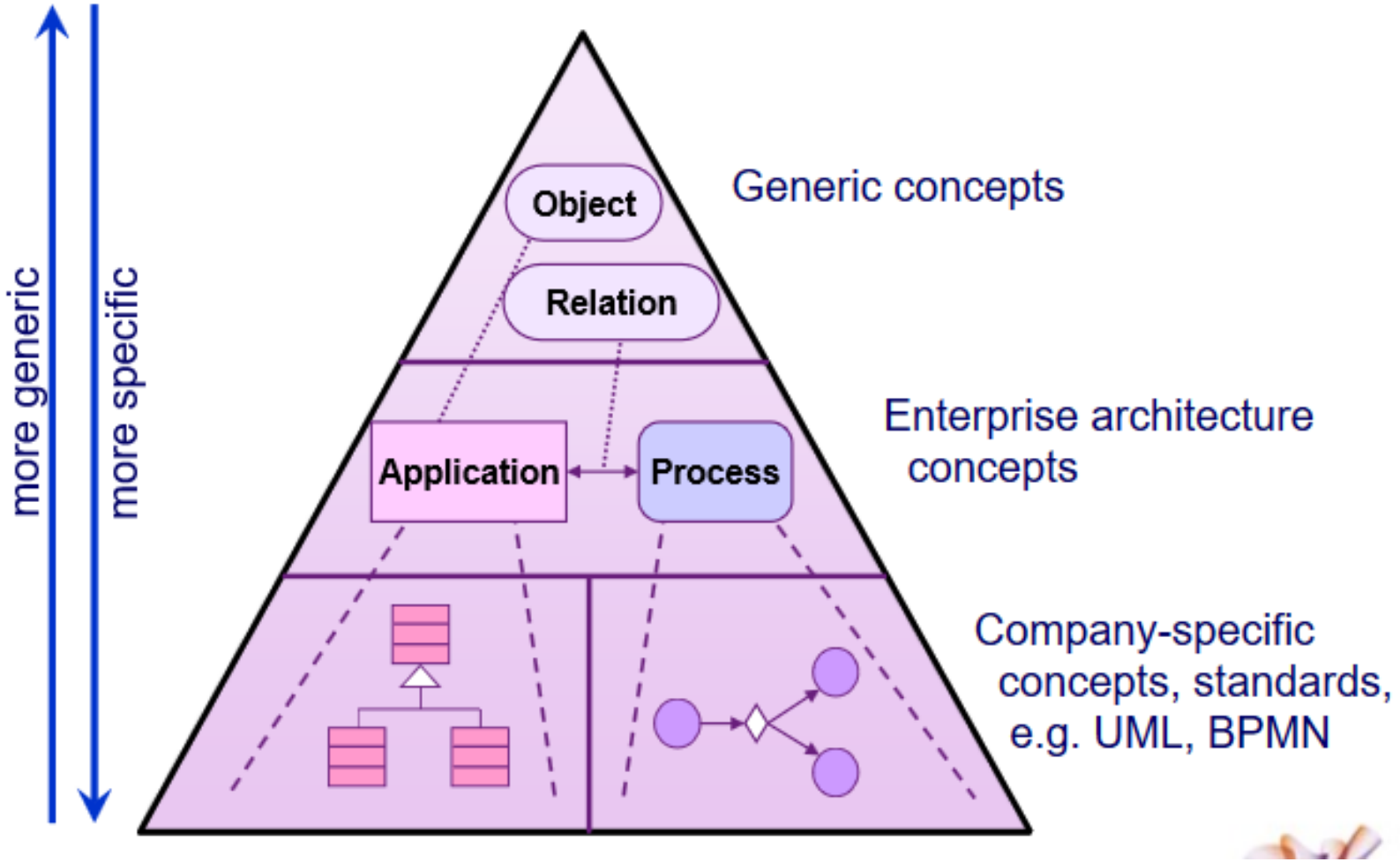
UML



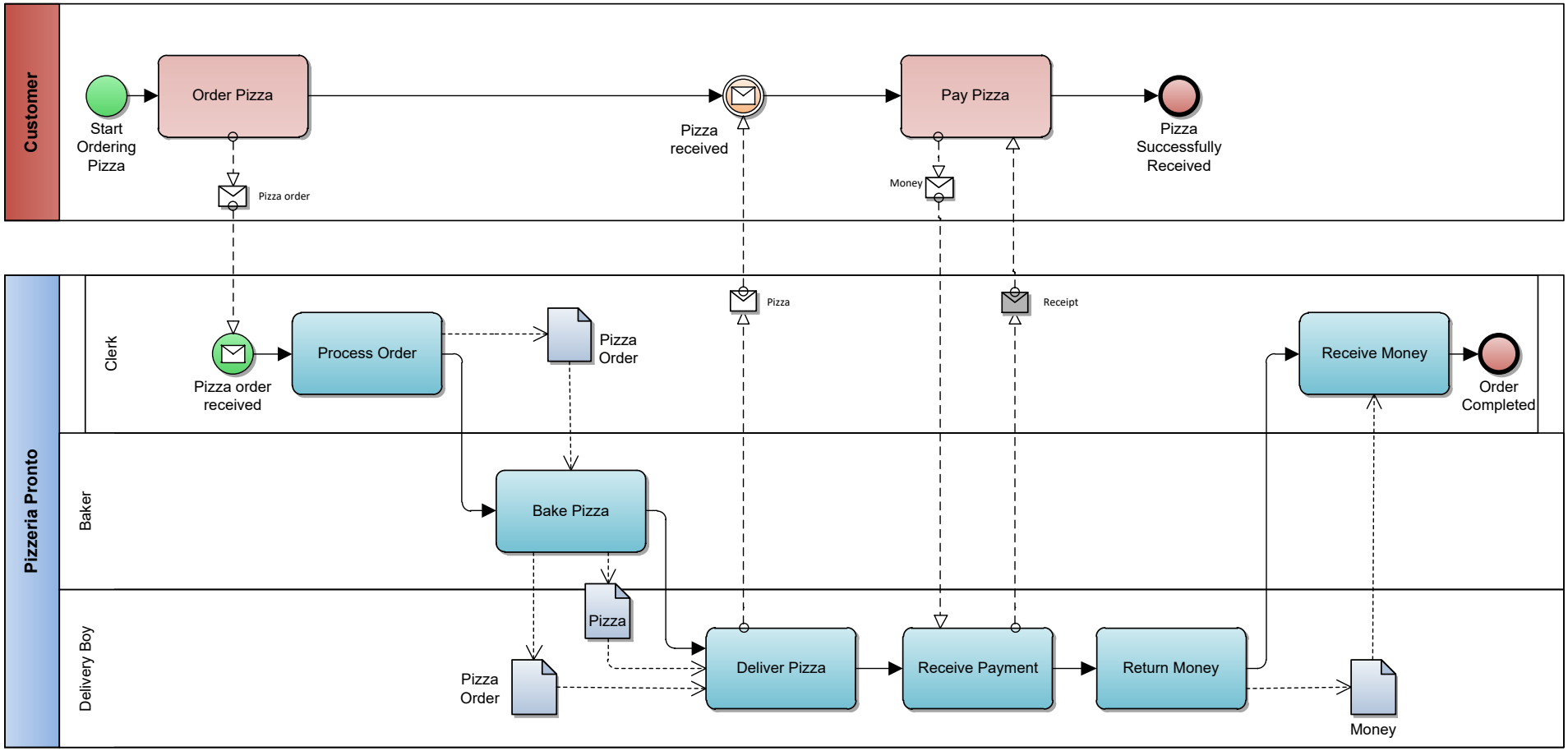
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http://slideplayer.com/slide/6051659/

Abstraction Levels



Pizzeria Process BPMN Collaboration Diagram



This simple collaboration shows the *sequence flows* (between the *lanes* of a *participant*), *message flows* (between *participants*) and the *data objects* and their *associations* to *activities* (i.e. the data flow). *Events* are used to synchronize the two *participants*. Each *message flow* shows what *message* is being sent or received.



About “Logical” and “Physical” Systems

- **“Logical” systems** are typically realized as software artefacts.
  - Focus on “**concept**” representation.
  - These concepts can be modelled using object orientation principles such as abstraction, classification and generalization.
  - **UML** specializes in modelling these systems.

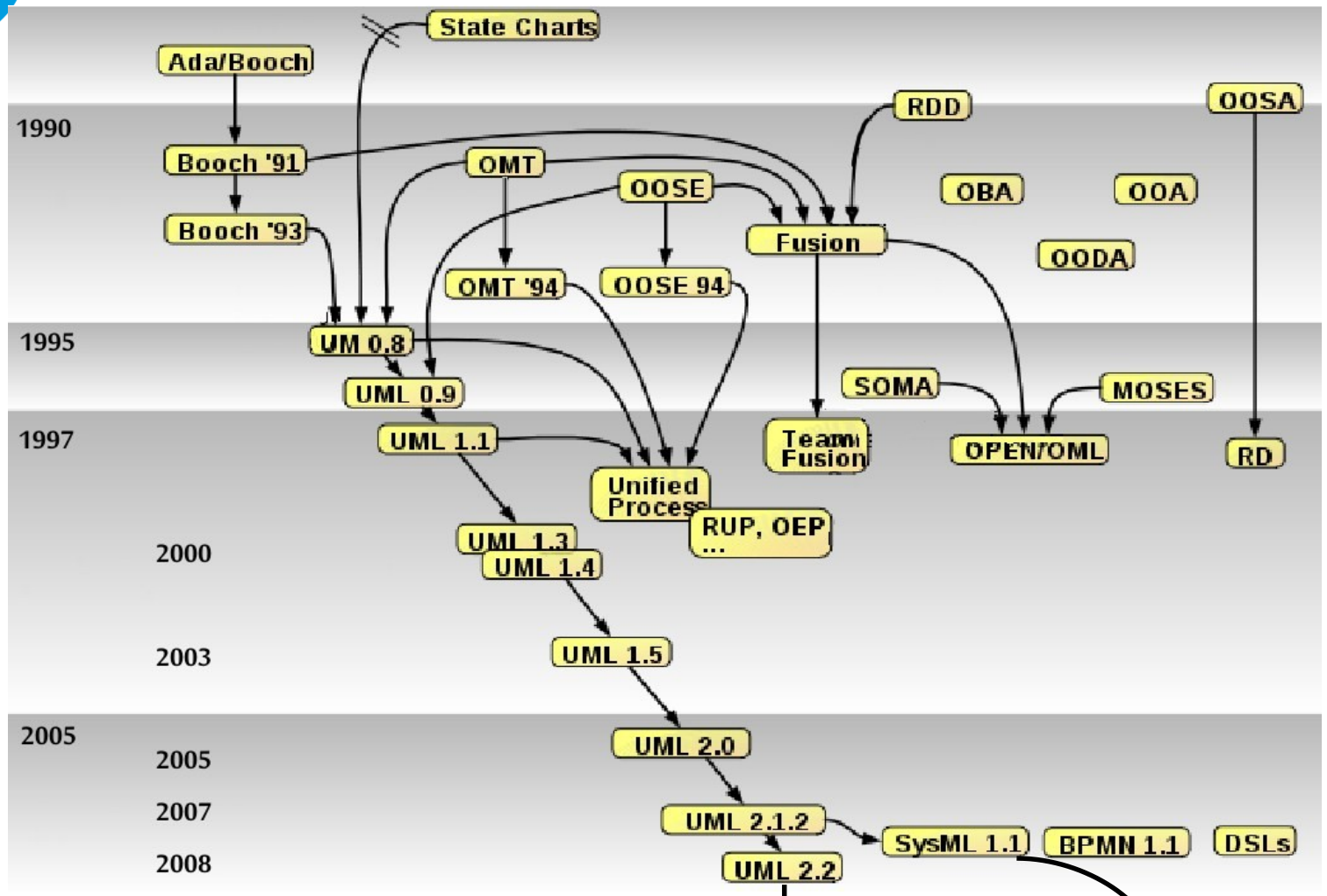
- **“Physical” systems** are typically realized as tangible artefacts, such as devices, machines and hardware.
  - Focus on **“object/part”** representation alongside with concepts.
  - **SysML** specializes in modelling these systems.

Systems Modelling Language

- In January 2001 the International Council on Systems Engineering (INCOSE) adopted UML as a language for systems engineering applications.
- UML was then adapted and extended to systems engineering using UML Profiles.

- The **Systems Modelling Language (SysML 1.0)** was approved by the OMG in July 2006
  - based on UML 2.1.1
- Current version is **SysML 1.3** (approved on June 2012)
  - based on UML 2.4

The path towards the UML and SysML:



What is “Enterprise Architecture”?

- **Enterprise architecture (EA)** is a discipline for proactively and holistically leading enterprise responses to disruptive forces by identifying and analyzing the execution of change toward desired business vision and outcomes. EA delivers value by presenting business and IT leaders with signature-ready recommendations for adjusting policies and projects to achieve target business outcomes that capitalize on relevant business disruptions.

http://www.gartner.com/it-glossary/enterprise-architecture-ea/

What is “Enterprise Architecture”?

- Enterprise Architecture (EA) is a method and an organizing principle that aligns functional business objectives and strategies with an IT strategy and execution plan. The Enterprise Architecture provides a guide to direct the evolution and transformation of enterprises with technology. This in turn makes IT a more strategic asset for successfully implementing a modern business strategy.
- An Enterprise Architecture typically produces deliverables such as:
  - Current State Enterprise Architecture model
  - Future State Enterprise Architecture reference model that is needed to execute on the proposed business strategy
  - Gap analysis that identifies the shortfalls of the current state in terms of its ability to support the objectives and strategies of the business
  - Architecture Roadmap that defines the initiatives required to migrate from the current stat into the future state.
- By taking an enterprise-wide perspective across business services, business processes, information, applications, and technology, an EA ensures the enterprise goals and objectives are addressed in a holistic way across all IT projects.

An Oracle White Paper in Enterprise Architecture
October 2009
The Oracle Enterprise Architecture Framework

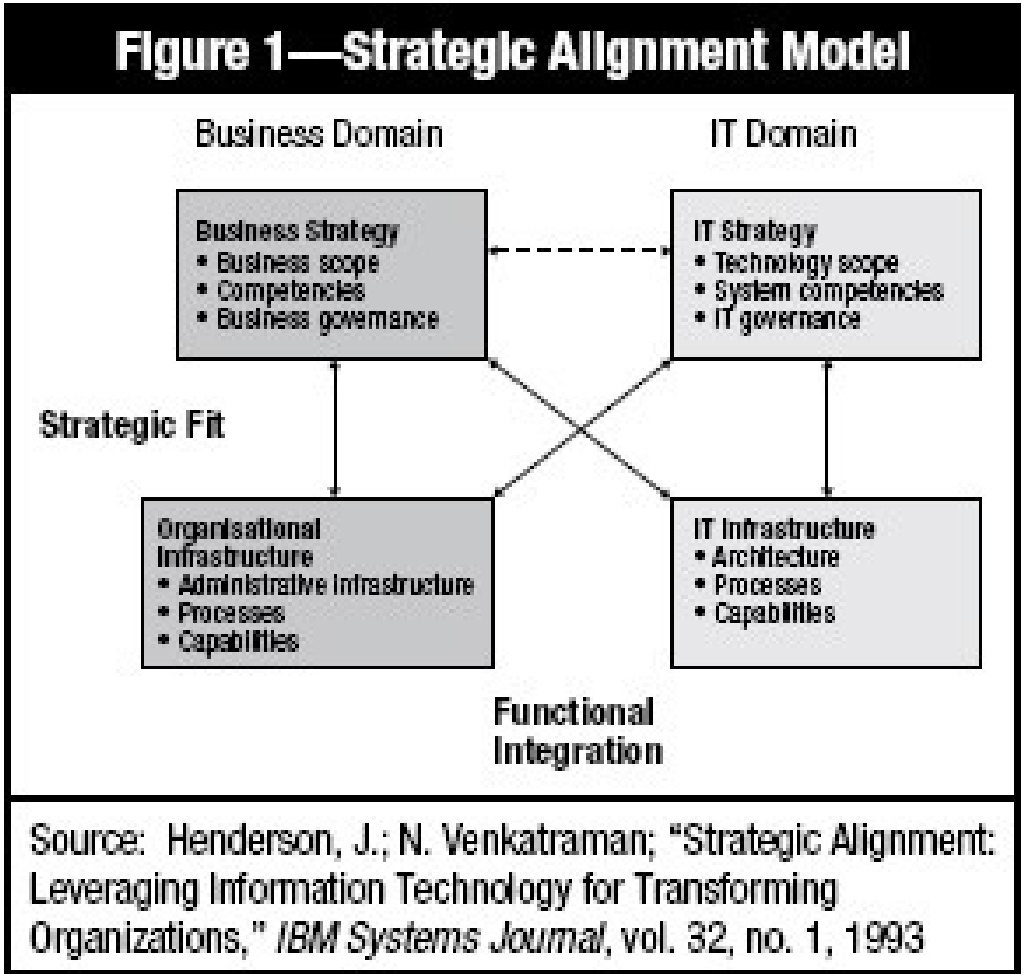
What is “Enterprise Architecture”?

- **TOGAF defines "enterprise" as any collection of organizations that has a common set of goals. For example, an enterprise could be a government agency, a whole corporation, a division of a corporation, a single department, or a chain of geographically distant organizations linked together by common ownership**.
- (…)
- The purpose of enterprise architecture is to optimize across the enterprise the often fragmented legacy of processes (both manual and automated) into an integrated environment that is responsive to change and supportive of the delivery of the business strategy.
- (…)
- Furthermore, a good enterprise architecture enables you to achieve the right balance between IT efficiency and business innovation.

http://pubs.opengroup.org/architecture/togaf9-doc/arch/

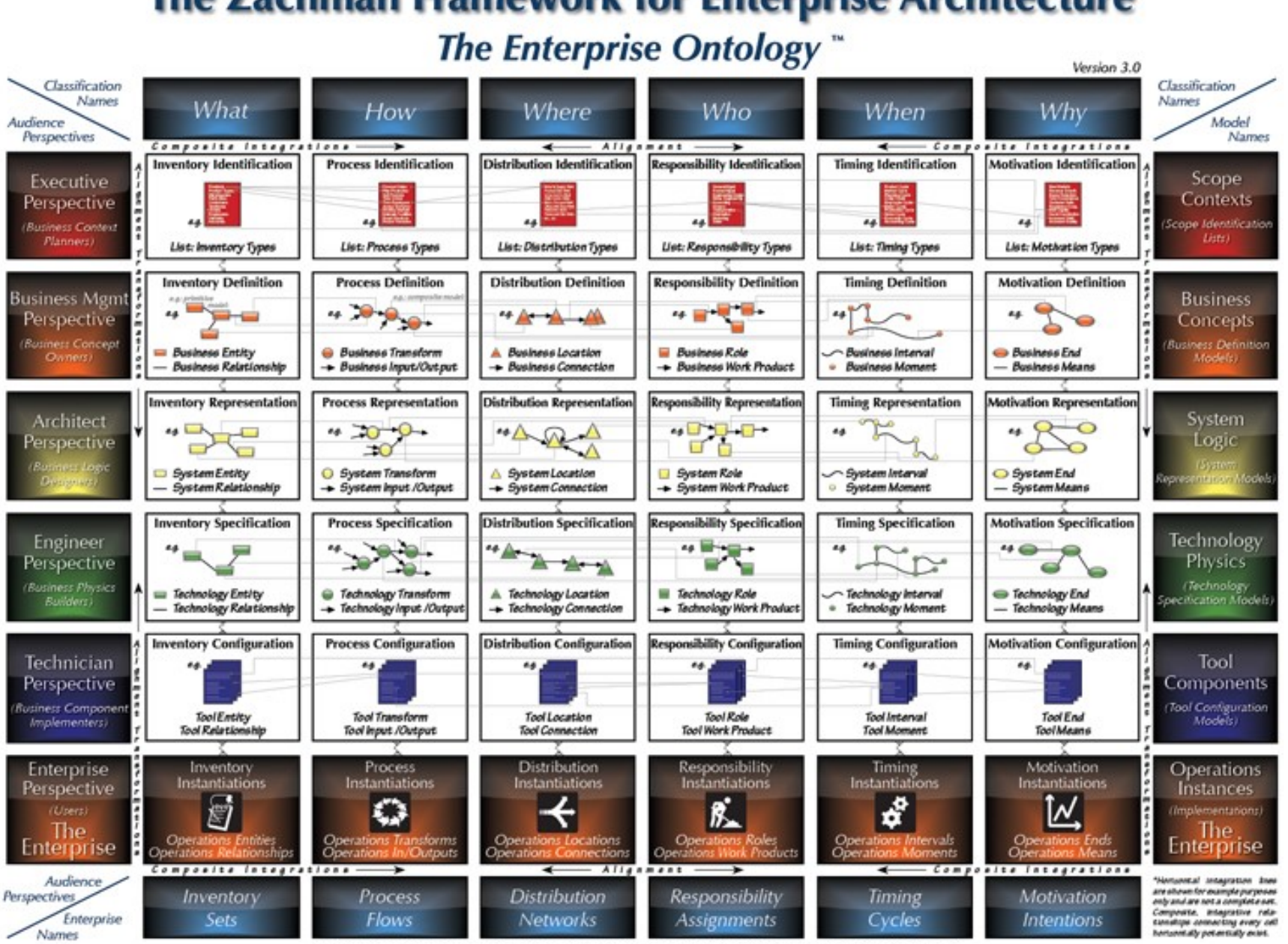
What is “Enterprise Architecture”?

In simple terms, EA is about **managing the Business versus IT Aligment**



https://www.zachman.com/

The *Zachman Framework™* is a schema - the intersection between two historical classifications that have been in use for literally thousands of years. The first is the fundamentals of communication found in the primitive interrogatives: **What, How, When, Who, Where, and Why**. It is the integration of answers to these questions that enables the comprehensive, composite description of complex ideas. The second is derived from reification, the transformation of an abstract idea into an instantiation that was initially postulated by ancient Greek philosophers and is labeled in the *Zachman Framework*™: **Identification, Definition, Representation, Specification, Configuration and Instantiation**.



https://www.zachman.com/ea-articles-reference/54-the-zachman-framework-evolution

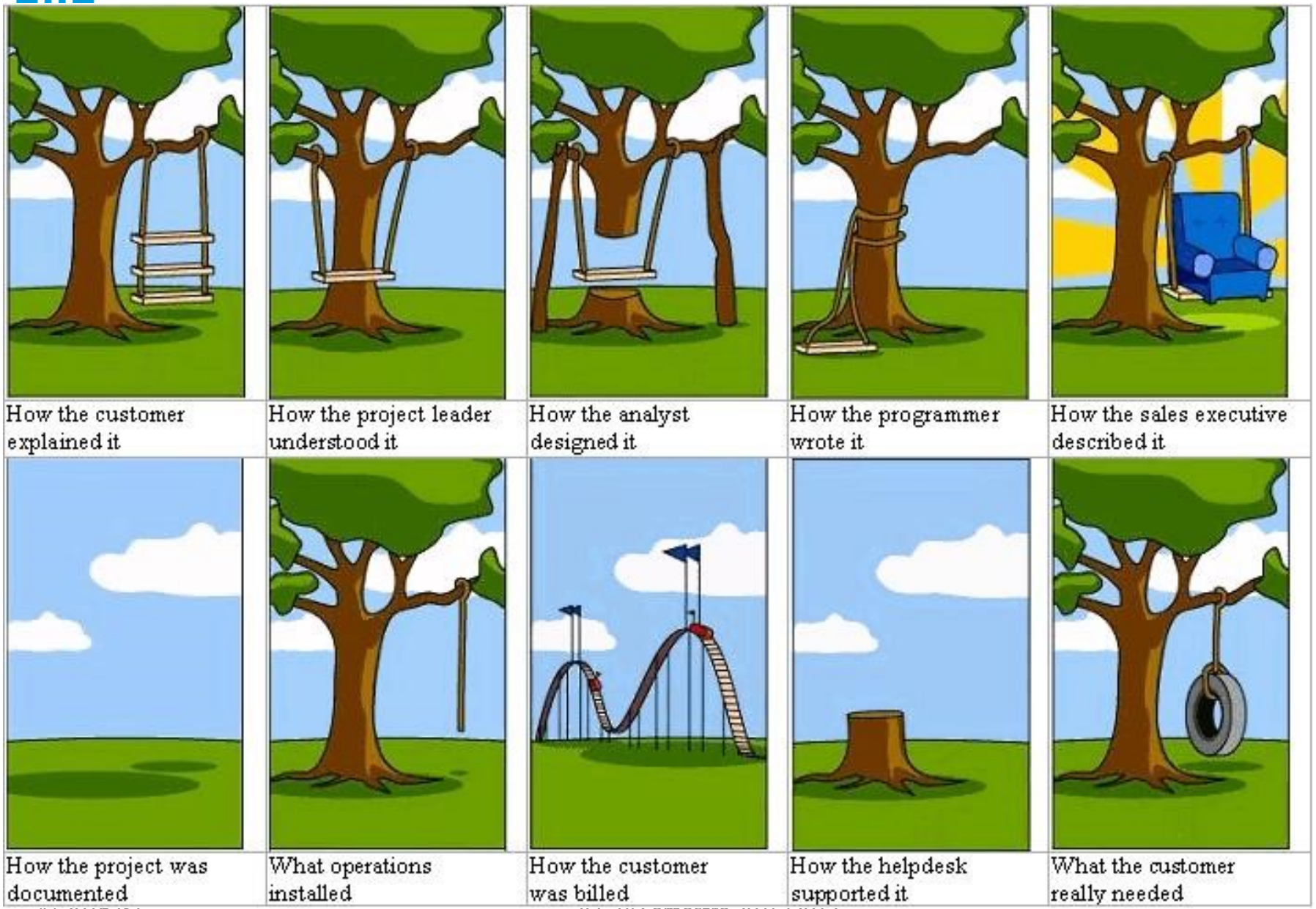
What is “Enterprise Architecture”?

	DATA *What*	FUNCTION *How*	NETWORK *Where*	PEOPLE *Who*	TIME *When*	MOTIVATION *Why*
Objective/Scope (contextual) *Role: Planner*	List of things important in the bu[illegible]ness	List of Business [illegible]Processes	List of Business [illegible]Locations	List of important [illegible]Organizations	List of [illegible]	List of Business Goal & [illegible]ategies
Enterprise Model (conceptual) *Role: Owner*	Conce[illegible]tual Data/ Objec[illegible]Model	Business Process Model	Business Logistics System	Work Flow Model	[illegible]Master Schedule	Bus[illegible]ness Pla[illegible]
System Model (logical) *Role:Designer*	Logical Data Model	System Architecture Model	Distributed Systems Architecture	Human Interface Architecture	Processing Structure	Business Rule Model
Technology Model (physical) *Role:Builder*	Physical Data/Class Model	Technology Design Model	Technology Architecture	Presentation Architecture	Control Structure	Rule Design
Detailed Reprentation (out of context) *Role: Programmer*	Data Definition	Program	Network Architecture	Security Architecture	Timing Definition	Rule Speculation
Functioning Enterprise *Role: User*	Usable Data	Working Function	Usable Network	Functioning Organization	Implemented Schedule	Working Strategy

ArchiMate

BPMN

UML


How the customer explained it
How the project leader understood it
How the analyst designed it
How the programmer wrote it
How the sales executive described it
How the project was documented
What operations installed
How the customer was billed
How the helpdesk supported it
What the customer really needed